

Learning from Random Samples

Ignorance gives one a large range of probabilities.

—GEORGE ELIOT

In Part I, we introduced the concept of random variables, and discussed their properties as a theoretical construct in the context of probability theory. Having established these fundamental concepts, in this chapter, we turn from mathematical probability theory to a canonical motivation of statistics: learning about features of *populations* through random sampling. We first consider the *independent and identically distributed (i.i.d.)* assumption and its ties to random sampling from a population (Section 3.1).

The i.i.d. assumption, if imposed, provides us with a great deal of inferential leverage. We demonstrate how we can estimate features of a random variable and, therefore, features of the population that gives rise to it under random sampling. We pay particular attention to the operating characteristics of estimators as the number of observations grows large (Section 3.2). Furthermore, we show how our degree of uncertainty about these estimates can *itself* be estimated by the same means, and that we can sometimes make probabilistic statements about the statistical uncertainty of a result (Section 3.4). Finally, we consider estimation and uncertainty when there is dependence in the data that can be characterized by discrete clusters (Section 3.5).

3.1 I.I.D. RANDOM VARIABLES

Our treatment of estimation in this book is based on the assumption that the researcher's data consist of *i.i.d.* observations of random variables or vectors. We begin with a formal definition.

Definition 3.1.1. *Independent and Identically Distributed (I.I.D.)*

Let $X_1, X_2, \dots, X_n$ be random variables with CDFs $F_1, F_2, \dots, F_n$, respectively. Let F_A denote the joint CDF of the random variables with indices in the set A . Then $X_1, X_2, \dots, X_n$ are *independent and identically distributed* if they satisfy the following:

- Mutually independent: $\forall A \subseteq \{1, 2, \dots, n\}, \forall (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$,

$$F_A((x_i)_{i \in A}) = \prod_{i \in A} F_i(x_i).^1$$
- Identically distributed: $\forall i, j \in \{1, 2, \dots, n\}$ and $\forall x \in \mathbb{R}, F_i(x) = F_j(x)$.

In other words, we take a draw of the random variable X . Then we take another draw of X in such a manner that our second draw does not depend on the outcome of the first. We repeat this process until we have n draws. So, we have n identical random variables that generate n values. We can subscript each of these n random variables, as shown above, as well as aggregate them to form the random vector $\mathbf{X} = (X_1, \dots, X_n)^T$.² Under the assumption of i.i.d. sampling from X , the collection of values that we see are assumed to be produced by an identical random process. When we have i.i.d. random variables $X_1, X_2, \dots, X_n$, we use the unsubscripted letter X to denote the random variable whose distribution the i.i.d. draws all share. So, for example, $E[X]$ denotes the expected value common to $X_1, X_2, \dots, X_n$, that is, $E[X] = E[X_1] = E[X_2] = \dots = E[X_n]$.³

3.1.1 Random Sampling

We proceed by considering a canonical motivation for the i.i.d. representation of uncertainty: tying the features of random variables to analogous features of populations through the process of *random sampling*. Suppose we have a *finite population* U consisting of N units, indexed $i = 1, 2, \dots, N$. The units of analysis may be, for example, people, villages, or countries. Associated with each unit i is a response x_i .⁴ We can describe the *finite population mass function*⁵ for the responses associated with the finite population U .

¹ Mutual independence implies that all observations are pairwise independent: $\forall i \neq j, X_i \perp X_j$, but the converse does not generally hold.

² We are expressing the random vector induced by random sampling as a column vector to clarify that each row represents a separate observation. Later, when we address i.i.d. draws of a random vector, this notational distinction will be helpful.

³ We will sometimes use subscripted variables inside the operator (for example, $E[X_i]$) when this notation improves clarity.

⁴ Nothing changes when we have *two* or more features associated with each unit. Suppose we had an ordered pair (x_i, y_i) associated with each unit i . When we randomly draw unit i , we observe and record both values in the pair.

⁵ This is not a standard term in statistics.

Definition 3.1.2. *Finite Population Mass Function*

Given a finite population U with responses $x_1, x_2, \dots, x_N$, the *finite population mass function*,

$$f_{FP}(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{I}(x_i = x).$$

That is, $f_{FP}(x)$ is the proportion of units in U that have $x_i = x$. Let $\mathcal{X}$ denote the set of unique values of x_i , that is, $\mathcal{X} = \{x \in \mathbb{R} : f_{FP}(x) > 0\}$. Let μ denote the population mean of U :

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i = \sum_{x \in \mathcal{X}} x f_{FP}(x),$$

and let σ^2 denote the population variance of U :

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \sum_{x \in \mathcal{X}} (x - \mu)^2 f_{FP}(x).$$

It will be helpful to have a concrete example to ground these ideas.

Example 3.1.3. *Finite Population Random Sampling* Suppose we have a finite population U consisting of $N = 4$ units. Let $x_1 = 3$, $x_2 = 4$, $x_3 = 3$, and $x_4 = 10$. Then

- $\mathcal{X} = \{3, 4, 10\}$.
- $\mu = \frac{1}{4}(3 + 4 + 3 + 10) = 5$.
- $\sigma^2 = \frac{1}{4}(4 + 1 + 4 + 25) = \frac{17}{2}$.
- $f_{FP}(-1) = 0$.
- $f_{FP}(3) = \frac{1}{2}$.
- $f_{FP}(4) = \frac{1}{4}$.
- $f_{FP}(10) = \frac{1}{4}$.
- $f_{FP}(24603) = 0$.
- $\forall x \notin \{3, 4, 10\}, f_{FP}(x) = 0$.

Let us consider a random generative process that selects one unit from U at random (with all units having equal probability of being selected). Let the random variable X take on the value of x_i associated with the unit selected.⁶ Each unit is selected with probability $\frac{1}{4}$, so we get $X = 3$ with probability $\frac{2}{4} = \frac{1}{2}$, $X = 4$ with probability $\frac{1}{4}$, and $X = 10$ with probability $\frac{1}{4}$. Note that this is simply a probability mass function. Under random sampling, the distribution

⁶ Formally, $\Omega = \{1, 2, 3, 4\}$; $P(\omega) = \frac{1}{4}$, $\forall \omega \in \Omega$; and $X(\omega) = x_\omega$, $\forall \omega \in \Omega$.

of outcomes in the population entirely determines the probability distribution of the random variable. The PMF of X is the finite population mass function: $f(x) = f_{FP}(x)$, $\forall x \in \mathbb{R}$, and $\mathcal{X} = \text{Supp}[X]$. Thus,

$$\begin{aligned} \bullet \quad E[X] &= \sum_x x f(x) = \sum_{x \in \mathcal{X}} x f_{FP}(x) = \mu = 5. \\ \bullet \quad V[X] &= E[(x - E[X])^2] = \sum_x (x - E[X])^2 f(x) = \sum_{x \in \mathcal{X}} (x - \mu)^2 f_{FP}(x) = \\ &\sigma^2 = \frac{17}{2}. \triangle \end{aligned}$$

So, under random sampling from U ,

- $E[X]$ is the *population mean*, that is, the average of all the x_i s.
- $V[X]$ is the *population variance*, that is, the average squared deviation from the mean of all the x_i s.

This allows us to cleanly identify the target of inquiry with respect to a particular set of N units. Consider two types of sampling schemes. The first is “with replacement” sampling: we “return” each unit after we draw it, so that the act of taking a draw does not change the population (and hence the population distribution). Thus, every draw is taken from the same distribution, independently and with its distribution governed by f_{FP} . In the case of “without replacement” sampling from a population, we can only observe each unit once and each draw affects the remaining population. However, if our population is sufficiently large, removing a single unit or a few units changes the distribution of the remaining units by a negligible amount. For example, suppose we had a population of 1,000,000 marbles consisting of 500,000 black marbles and 500,000 white marbles. Suppose we sample n marbles without replacement. Let $X_i = 0$ if marble i in our sample is black and $X_i = 1$ if marble i is white. Then $\Pr[X_2 = 1] = \frac{500,000}{1,000,000} = 0.5$ and $\Pr[X_2 = 1 | X_1 = 1] = \frac{499,999}{999,999} \approx 0.5$. Similarly, the probability that marble n is black is approximately 0.5 regardless of the colors of marbles 1 through $n - 1$, as long as n is a very small fraction of 1,000,000. Some technical finesse is required to formalize this intuition (e.g., Dubins and Freedman 1979), but it implies that sampling units at random from a hypothetical, infinite *superpopulation* yields data that can be treated as i.i.d.

3.1.2 I.I.D. as Approximation

The i.i.d. assumption need not reflect the manner in which social scientific data are collected in order for it to be a useful approximation. A literal interpretation of the above section as a motivation for the i.i.d. assumption would leave us with only very limited applications for statistical inference. However, for arguments to proceed, frameworks must be set, and the i.i.d. assumption provides one fairly minimal way to do so. The communication of empirical findings often requires some way in which frameworks for inference can be

well defined. Accordingly, the superpopulation model can be viewed as a codification of uncertainty about generalizability to broader populations, through the above analogy of random sampling.⁷

The plausibility of the i.i.d. assumption as an approximation depends largely on two elements, namely that the data are approximately *independent* and also approximately *identically distributed*. We first consider settings where the data may not be independent. That is, there may be sources of dependence between units: for example, units 1 and 2 might be intertwined (because they are, say, members of the same household) and so the assumption that $Y_1 \perp\!\!\!\perp Y_2$ would not be a reasonable way to characterize the units under study. In such settings, approaches that erroneously assume the i.i.d. assumption will not generally yield valid inferences. But, assuming that this dependence is not too great between units (loosely speaking, there is relevant information added by new units as we grow in n), then our results will largely carry through—with the caveat that we will be likely to understate uncertainty. Section 3.5 considers one special case, the case where dependence can be represented by discrete clusters, and illustrates some of the consequences of such clustering.

Similarly, we might worry that the data are not identically distributed. When this occurs, the i^{th} observation may have a different distribution than the j^{th} observation. This occurs, for example, when observations are collected over time, and there are changes to the population under study over time. (For example, if collecting survey evidence, opinions may change over time.) In such cases, we are no longer able to neatly describe the distribution of each random variable in terms of a single population distribution. As a statistical matter, most of our results in this chapter will carry through with little modification even if we assume independent but *not* identically distributed data. The primary consequence will lie with interpretation. Instead of being able to make clean statements about features of a single population, our statements are instead about a *pseudopopulation* that aggregates over the different distributions associated with each unit. However, this aggregate pseudopopulation is a moving target, and our statistical claims are circumscribed accordingly. When the data are not identically distributed, predictions for the $(n + 1)^{\text{th}}$ unit

⁷ Though we do not fully endorse its contents, we are fond of the following quotation from Angrist (2013):

Some would say all data come from “super-populations,” that is, the data we happen to have could have come from other times, other places, or other people, even if we seem to [observe] everyone from a particular scenario. Others take a model-based approach: some kind of stochastic process generates the data at hand; there is always more where they came from. Finally, an approach known as randomization inference recognizes that even in finite populations, counterfactuals remain hidden, and therefore we always require inference. You could spend your life pondering such things. I have to admit I try not to.

generated using the first n units have few guarantees. But as a means for making statements about a (pseudo)population that the data at hand represent, the i.i.d. assumption may nevertheless be a suitable approximation.

3.2 ESTIMATION

If we observed just one draw of a random variable X , we probably could not learn too much about its distribution. Unless X had no variance, any guess about a feature of its distribution (such as its CDF or expected value) would be subject to a great deal of chance variability. We would not be able to *estimate* a feature of the distribution of X with much reliability.

But what if we observed multiple draws of a random variable? As our sample, or collection of observations from the same random variable X , grows larger and larger, we are able to estimate the features of X with more and more precision. We begin here with a rather general definition of a way to summarize the observed data. A *sample statistic* summarizes the values of $X_1, X_2, \dots, X_n$.

Definition 3.2.1. Sample Statistic

For i.i.d. random variables $X_1, X_2, \dots, X_n$, a *sample statistic* is a function of $X_1, X_2, \dots, X_n$:

$$T_{(n)} = h_{(n)}(X_1, X_2, \dots, X_n),$$

where $h_{(n)} : \mathbb{R}^n \rightarrow \mathbb{R}, \forall n \in \mathbb{N}$.

Importantly, since a sample statistic is a function of the random variables $X_1, X_2, \dots, X_n$, it is therefore *itself* a random variable. Furthermore, a sample statistic defines a sequence of random variables indexed by the sample size n . (We use parenthetic subscripts to index elements of a sequence, though we often drop the subscript when discussing sample statistics.) This will be important when we discuss the behavior of sample statistics as $n \rightarrow \infty$.

3.2.1 Sample Means

We will now focus on the properties of one foundational sample statistic: the sample mean. As we will see in the following sections, many of the intuitions in the case of the sample mean will generalize to other sample statistics. The sample mean of n draws from X can be viewed as an approximation of $E[X]$. It is the average of all the observed values of X .

Definition 3.2.2. Sample Mean

For i.i.d. random variables $X_1, X_2, \dots, X_n$, the *sample mean* is

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Keep in mind that the sample mean is a sample statistic and thus is itself a random variable. Depending on the outcomes of the random variables $X_1, X_2, \dots, X_n$, $\bar{X}$ will take on different values.

We now derive some important properties of $\bar{X}$. First, we will show that the sample mean tends to approximate the population mean, in the sense that “on average” (across hypothetical realizations of the values of $X_1, X_2, \dots, X_n$) the value of $\bar{X}$ is $E[X]$.

Theorem 3.2.3. The Expected Value of the Sample Mean Is the Population Mean

For i.i.d. random variables $X_1, X_2, \dots, X_n$,

$$E[\bar{X}] = E[X].$$

Proof:

$$\begin{aligned} E[\bar{X}] &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n}E[X_1 + X_2 + \dots + X_n] \\ &= \frac{1}{n}(E[X_1] + E[X_2] + \dots + E[X_n]) \\ &= \frac{1}{n}(E[X] + E[X] + \dots + E[X]) \\ &= \frac{1}{n}nE[X] \\ &= E[X]. \quad \square \end{aligned}$$

As $\bar{X}$ is a random variable, we can describe other features of its distribution. The variance of $\bar{X}$ is known as the *sampling variance* of the sample mean. Much as the variance characterizes the variability of a random variable X , the sampling variance of the sample mean characterizes how much variability we can expect in the sample mean across hypothetical draws of n observations.

Theorem 3.2.4. Sampling Variance of the Sample Mean

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite variance $V[X]$, the sampling variance of $\bar{X}$ is

$$V[\bar{X}] = \frac{V[X]}{n}.$$

Proof:

$$\begin{aligned} V[\bar{X}] &= V\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] \\ &= \frac{1}{n^2} V[X_1 + X_2 + \dots + X_n] \\ &= \frac{1}{n^2} (V[X_1] + V[X_2] + \dots + V[X_n]) \\ &= \frac{1}{n^2} (V[X] + V[X] + \dots + V[X]) \\ &= \frac{1}{n^2} nV[X] \\ &= \frac{V[X]}{n}. \quad \square \end{aligned}$$

Notice that the sampling variance $V[\bar{X}]$ decreases as n increases. This fact, along with Theorem 3.2.3, underlies an important result that we will present shortly. First, however, we must establish the following theorem, which bounds the probability that the sample mean is far from the population mean.

Theorem 3.2.5. Chebyshev's Inequality for the Sample Mean

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite variance $V[X] > 0$. Then, $\forall \epsilon > 0$,

$$\Pr\left[|\bar{X} - E[X]| \geq \epsilon\right] \leq \frac{V[X]}{\epsilon^2 n}.$$

Proof: This follows directly from the previous two theorems and Chebyshev's Inequality (Theorem 2.1.18). Let $\epsilon > 0$ and let $\epsilon' = \frac{\epsilon}{\sigma[\bar{X}]}$, so that $\epsilon = \epsilon' \sigma[\bar{X}]$. Then

$$\begin{aligned} \Pr\left[|\bar{X} - E[X]| \geq \epsilon\right] &= \Pr\left[|\bar{X} - E[\bar{X}]| \geq \epsilon' \sigma[\bar{X}]\right] \\ &\leq \frac{1}{\epsilon'^2} \\ &= \frac{\sigma[\bar{X}]^2}{\epsilon^2} \end{aligned}$$

$$\begin{aligned}
 &= \frac{V[\bar{X}]}{\epsilon^2} \\
 &= \frac{V[X]}{\epsilon^2 n}. \quad \square
 \end{aligned}$$

Theorem 3.2.5 will allow us to prove a crucial theorem in statistics, the *Weak Law of Large Numbers (WLLN)*. To state the WLLN concisely, we will also require the following definition.

Definition 3.2.6. Convergence in Probability

Let $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be a sequence of random variables and let $c \in \mathbb{R}$. Then $T_{(n)}$ *converges in probability to c* if, $\forall \epsilon > 0$,

$$\lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| \geq \epsilon] = 0,$$

or equivalently,⁸

$$\lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| < \epsilon] = 1.$$

We write $T_{(n)} \xrightarrow{p} c$ to denote that $T_{(n)}$ converges in probability to c .⁹

The value c to which a sequence of random variables $T_{(n)}$ converges in probability is referred to as the *probability limit* of $T_{(n)}$. Saying that $T_{(n)}$ converges in probability to c means that as n gets large, it becomes increasingly likely that $T_{(n)}$ will be “close” to c . More specifically, for any $p \in [0, 1]$ and any $\epsilon > 0$, there is always an N large enough that, for every $n \geq N$, the probability that $T_{(n)}$ will be within ϵ of c is at least p . As $n \rightarrow \infty$, it becomes extremely likely that $T_{(n)}$ will be extremely close to c . This definition can apply to any sequence of random variables, but we use $T_{(n)}$ suggestively, as we will generally discuss convergence in probability of sample statistics.

We briefly digress here to note that convergence in probability of a sequence $T_{(n)}$ implies that any continuous function of $T_{(n)}$ (and potentially other sequences that converge in probability) itself converges in probability to the

⁸ The equivalence follows from the complement rule (see Theorem 1.1.4, *Basic Properties of Probability*):

$$\lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| < \epsilon] = \lim_{n \rightarrow \infty} \left(1 - \Pr[|T_{(n)} - c| \geq \epsilon]\right) = 1 - \lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| \geq \epsilon],$$

so

$$\lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| \geq \epsilon] = 0 \iff \lim_{n \rightarrow \infty} \Pr[|T_{(n)} - c| < \epsilon] = 1.$$

⁹ A more general definition allows for $T_{(n)}$ to converge in probability to a random variable T : for a sequence of random variables $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ and a random variable T , $T_{(n)} \xrightarrow{p} T$ if $\forall \epsilon > 0$, $\lim_{n \rightarrow \infty} \Pr[|T_{(n)} - T| \geq \epsilon] = 0$.

value of the function at the probability limit(s). This result is known as the *Continuous Mapping Theorem (CMT)*, which we state without proof.

Theorem 3.2.7. Continuous Mapping Theorem (CMT)

Let $(S_{(1)}, S_{(2)}, S_{(3)}, \dots)$ and $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be sequences of random variables. Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function and let $a, b \in \mathbb{R}$. If $S_{(n)} \xrightarrow{p} a$ and $T_{(n)} \xrightarrow{p} b$, then $g(S_{(n)}, T_{(n)}) \xrightarrow{p} g(a, b)$.

The CMT says that continuous functions preserve convergence in probability. Although Theorem 3.2.7 states the CMT for the case of functions of two random variables, it also applies to functions of a single random variable and functions of more than two random variables. While we will not be using it immediately, the CMT will be useful when we seek to generalize to sample statistics other than the sample mean, and we will invoke it frequently in proofs.

With our definition of convergence in probability, we can now state the Weak Law of Large Numbers.

Theorem 3.2.8. Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite variance $V[X] > 0$,¹⁰ and let $\bar{X}_{(n)} = \frac{1}{n} \sum_{i=1}^n X_i$. Then

$$\bar{X}_{(n)} \xrightarrow{p} E[X].$$

Proof: By non-negativity of probability and Theorem 3.2.5, $\forall \epsilon > 0$,

$$0 \leq \Pr \left[|\bar{X}_{(n)} - E[X]| \geq \epsilon \right] \leq \frac{V[X]}{\epsilon^2 n},$$

and

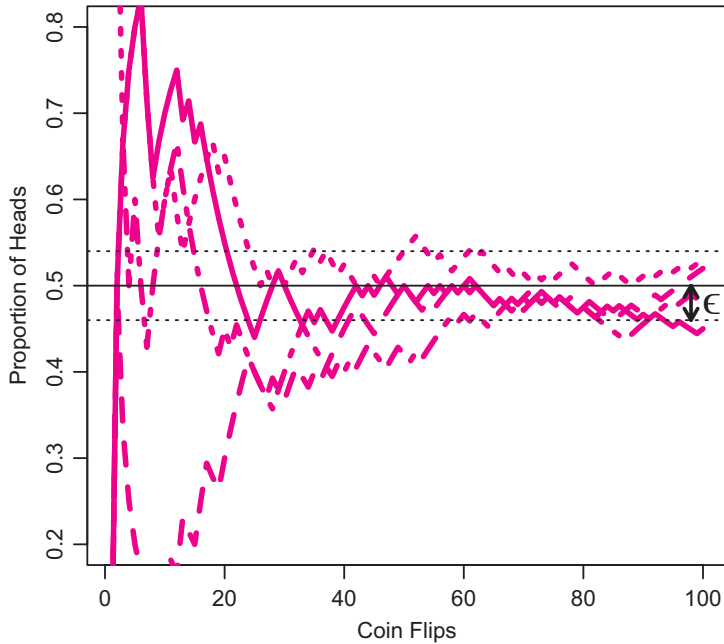
$$\lim_{n \rightarrow \infty} 0 = \lim_{n \rightarrow \infty} \frac{V[X]}{\epsilon^2 n} = 0,$$

so by the Squeeze Theorem for Sequences,¹¹

$$\lim_{n \rightarrow \infty} \Pr \left[|\bar{X}_{(n)} - E[X]| \geq \epsilon \right] = 0. \quad \square$$

¹⁰ Note that if $V[X] = 0$, the result follows trivially.

¹¹ Squeeze Theorem for Sequences: if $\lim_{n \rightarrow \infty} b_{(n)} = \lim_{n \rightarrow \infty} c_{(n)} = L$ and there exists an integer N such that $\forall n \geq N, b_{(n)} \leq a_{(n)} \leq c_{(n)}$, then $\lim_{n \rightarrow \infty} a_{(n)} = L$.

FIGURE 3.2.1. *Weak Law of Large Numbers*

The WLLN is a profound result: as n gets large, the sample mean $\bar{X}$ becomes increasingly likely to approximate $E[X]$ to any arbitrary degree of precision. That is, with a large enough sample, the probability that the sample mean will be far from the population mean will be negligible. (Again, for notational simplicity, we will drop the parenthetical subscript on sample statistics, even when discussing their properties as $n \rightarrow \infty$. For example, we will write $\bar{X} \rightarrow E[X]$.)

Figure 3.2.1 shows a graphical representation of the WLLN. Each curve in the figure shows the running proportion of heads for a simulated sequence of coin flips. Notice how the cumulative proportion of heads tends to get closer and closer to 0.5. Indeed, for any given distance ϵ from 0.5 (represented by the horizontal dotted lines), there is some number of coin tosses after which it would be virtually certain that the running proportion of heads would be within ϵ of 0.5.

The generality of the WLLN is illustrated in the following theorem, which shows how the sample mean can be used to estimate the CDF at any point. Given that the CDF characterizes everything about the distribution of a random variable (Section 1.2.3, *Cumulative Distribution Functions*), this result is profound. It will also serve as the basis for our general discussion of plug-in estimation in Section 3.3 (*The Plug-In Principle*).

Theorem 3.2.9. Estimating the CDF

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with common CDF F . Let $x \in \mathbb{R}$ and let $Z_i = I(X_i \leq x)$, $\forall i \in \{1, 2, \dots, n\}$, where $I(\cdot)$ is the *indicator function*, that is, it takes the value one if its argument is true and zero if it is false.¹² Then

$$\bar{Z} \xrightarrow{p} F(x).$$

Proof: By a generalization of Theorem 1.3.17 (*Implications of Independence, Part I*), $Z_1, Z_2, \dots, Z_n$ are i.i.d. random variables with common PMF

$$f_Z(z) = \begin{cases} \Pr[X \leq x] & : z = 1 \\ \Pr[X > x] & : z = 0 \\ 0 & : \text{otherwise.} \end{cases}$$

So, by the definition of the expected value,

$$E[Z] = \sum_z z f_Z(z) = 1 \cdot \Pr[X \leq x] + 0 \cdot \Pr[X > x] = \Pr[X \leq x] = F(x).$$

Thus, by the WLLN,¹³

$$\bar{Z} \xrightarrow{p} E[Z] = F(x). \quad \square$$

Note that we could apply this theorem for any or every value of x . Thus, given a sufficient number of observations, we can always approximate the value of the CDF at any point to arbitrary precision. If n is very large, the probability that we are far off from the true value of $F(x)$ is very small.¹⁴ So, since the CDF tells us everything about a random variable, this means that we can estimate features of a random variable to arbitrary precision, given large enough n . We show how this works more generally in Section 3.3, *The Plug-In Principle*. First, though, we must define the general notion of an estimator.

3.2.2 Estimation Theory

Suppose there is some feature θ associated with random variable X . For example, θ could be the expected value of X , $E[X]$, or it could be something entirely odd, like $V[(X - E[X])^3]$. As usual, we observe a sample of n i.i.d. draws of X ,

¹² Note that for any random variable X , $\forall D \in \mathbb{R}$, $E[I(X \in D)] = \Pr[X \in D]$.

¹³ Technically, if $\Pr[X \leq x] = 0$ or 1 , then Z is a degenerate random variable, so $V[Z] = 0$ and the WLLN is not applicable. But in this case, the result holds trivially. Otherwise, Z has finite $E[Z]$ and finite $V[Z] > 0$, so the WLLN applies.

¹⁴ Stronger results are possible under weaker assumptions, namely the Strong Law of Large Numbers and the Glivenko–Cantelli Theorem.

which is equivalent to a single draw of the random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$. An *estimator* of θ is a random variable $\hat{\theta} = h(\mathbf{X}) = h(X_1, X_2, \dots, X_n)$, and the value it takes on is called an *estimate*.¹⁵ Thus, formally, an estimator is nothing more than a sample statistic (see Definition 3.2.1). But we would not refer to just any sample statistic as an estimator. We reserve the term for sample statistics that, in some sense, provide “good” estimates of some population feature θ .

One property often considered to make an estimator “good” is unbiasedness.

Definition 3.2.10. Unbiasedness

An estimator $\hat{\theta}$ is *unbiased* for θ if $E[\hat{\theta}] = \theta$.

An unbiased estimator gets the right answer “on average.” Theorem 3.2.3, *The Expected Value of the Sample Mean Is the Population Mean*, can thus be stated as follows: “The sample mean is unbiased for the population mean.”

An estimator that is not unbiased is, as one might expect, referred to as *biased*. We can quantify the bias of an estimator as follows:

Definition 3.2.11. Bias of an Estimator

For an estimator $\hat{\theta}$, the *bias* of $\hat{\theta}$ in estimating θ is $E[\hat{\theta}] - \theta$.

The bias tells us the difference between the “average” value of the estimator and the true value of θ . An unbiased estimator, naturally, has bias equal to zero. Unbiasedness is a good property for an estimator to have, though it is less important than it might initially seem. Unbiasedness alone says little about how well we can expect $\hat{\theta}$ to approximate θ . It does not guarantee that our estimate will usually, or even ever, come close to the true value.¹⁶

Our usual definitions that allow us to characterize random variables also allow us to characterize the distributions of estimators. (This, of course, is because estimators are themselves random variables.) Some of these properties

¹⁵ For some estimators that we will consider, there may be some positive probability that $X_1, X_2, \dots, X_n$ take on values such that $h(\mathbf{X}) = \frac{0}{0}$. In such cases, we will implicitly assume that when this happens, the estimator takes on the value of zero. For well-behaved estimators, this will occur with vanishingly small probability as n grows large.

¹⁶ The fixation on bias of many statisticians results in the following joke (a slightly modified version of Riesbeck N.d., itself a variation on an old joke among statisticians):

A physicist, an engineer, and a statistician are out hunting. Suddenly, a deer appears 50 yards away. The physicist does some basic ballistic calculations, assuming a vacuum, lifts her rifle to a specific angle, and shoots. The bullet lands 5 yards short. The engineer adds a fudge factor for air resistance, lifts his rifle slightly higher, and shoots. The bullet lands 5 yards long. The statistician yells, “We got ‘em!”

have special names in the context of estimators. For example, the variance of an estimator is called the *sampling variance*.

Definition 3.2.12. *Sampling Variance of an Estimator*

For an estimator $\hat{\theta}$, the *sampling variance* of $\hat{\theta}$ is $V[\hat{\theta}]$.

For example, as in Theorem 3.2.4, we would say that the sampling variance of the sample mean $\bar{X}$ is $\frac{V[X]}{n}$. The standard deviation of an estimator is similarly important enough to merit its own name: the *standard error*.

Definition 3.2.13. *Standard Error of an Estimator*

For an estimator $\hat{\theta}$, the *standard error* of $\hat{\theta}$ is $\sigma[\hat{\theta}]$.

The standard error has great utility as a metric of variability, and will be central to how we make statistical *inferences* (Section 3.4). Another similarly foundational concept is the *mean squared error* (MSE) of an estimator, or simply the mean squared error of $\hat{\theta}$ about θ .

Definition 3.2.14. *MSE of an Estimator*

For an estimator $\hat{\theta}$, the *mean squared error* (MSE) of $\hat{\theta}$ in estimating θ is $E[(\hat{\theta} - \theta)^2]$.

Attentive readers may notice that we have already defined MSE in Section 2.1.3. Indeed, this is the same MSE that we defined it in Definition 2.1.22, just applied to estimators. The MSE of an estimator is the expected squared deviation of the estimate from the true value. As before, we can also write MSE in an alternative form.

Theorem 3.2.15. *Alternative Formula for the MSE of an Estimator*

For an estimator $\hat{\theta}$,

$$E[(\hat{\theta} - \theta)^2] = V[\hat{\theta}] + (E[\hat{\theta}] - \theta)^2.$$

Proof: This follows directly from Theorem 2.1.23. $\square$

We can describe the meaning of this formula in words: the MSE of an estimator is equal to its sampling variance plus the square of its bias. We can thus see why unbiasedness alone is insufficient to ensure that an estimator is sensible: an estimator with zero bias but large sampling variance may often, or even always, be very far off from the true value of θ . All else equal, we prefer estimators with a lower MSE, as that implies that our estimates are better approximations. An estimator that has a lower MSE than another is said to be more *efficient*.

Definition 3.2.16. Relative Efficiency

Let $\hat{\theta}_A$ and $\hat{\theta}_B$ be estimators of θ . Then $\hat{\theta}_A$ is *more efficient* than $\hat{\theta}_B$ if it has a lower MSE.

We will not explicitly consider efficiency too much in this book, though we may occasionally make reference to the efficiency of one estimator relative to another; these comparisons allow us to evaluate the suitability of estimators for a given problem.

Note that the above properties all implicitly assume some finite sample size n . We can also describe the *asymptotic* properties of an estimator, that is, the properties of $\hat{\theta}$ as $n \rightarrow \infty$. The simplest and perhaps most essential of these properties is *consistency*.

Definition 3.2.17. Consistency

An estimator $\hat{\theta}$ is *consistent* for θ if $\hat{\theta} \xrightarrow{P} \theta$.

Consistency is a simple notion: if we had enough data, the probability that our estimate $\hat{\theta}$ would be far from the truth θ would be close to zero. The WLLN can thus be stated as follows: “The sample mean is consistent for the population mean.” If $\hat{\theta}$ was not in fact consistent, then even infinite data could not give us a high probability of approximating θ well. Consistency can be reasonably considered to be a minimum requirement for an estimator to qualify as reasonable.

Importantly, although unbiased estimators are not necessarily consistent, any unbiased estimator $\hat{\theta}$ with $\lim_{n \rightarrow \infty} V[\hat{\theta}] = 0$ is consistent.¹⁷ (This follows from Theorem 2.1.18, *Chebyshev’s Inequality*.) Thus, even when we do not necessarily value unbiasedness, it may be useful in establishing other properties of an estimator. We provide more details on the asymptotic properties of estimators in Section 3.2.5.

3.2.3 Variance Estimators

With these concepts in mind, we can now make our first attempt at proposing an estimator of $V[X]$, or the *population variance* of a random variable X . Recall from Theorem 2.1.13 (*Alternative Formula for Variance*) that $V[X] = E[X^2] - E[X]^2$. We know from the WLLN that the sample mean converges in probability to the expected value or population mean, so what if we “plug in” sample means for expected values? We will call this the *plug-in sample variance*.¹⁸

¹⁷ The converse, however, does not necessarily hold: an unbiased and consistent estimator may nonetheless have positive (or even infinite) sampling variance (and thus, MSE) even as $n \rightarrow \infty$.

¹⁸ Plug-in estimators can also be referred to as “analog estimators,” based on the “analogy principle of estimation,” which encompasses the general use of sample statistics as estimators

Definition 3.2.18. *Plug-In Sample Variance*

For i.i.d. random variables $X_1, X_2, \dots, X_n$, the *plug-in sample variance* is $\overline{X^2} - \overline{X}^2$.

Note that

$$\overline{X^2} = \frac{1}{n} \sum_{i=1}^n X_i^2,$$

that is, the sample mean of X^2 , and

$$\overline{X}^2 = \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2,$$

that is, the square of the sample mean of X .

The following theorem describes the key properties of the plug-in sample variance.

Theorem 3.2.19. *Properties of the Plug-In Sample Variance*

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite variance $V[X] > 0$,

- $E[\overline{X^2} - \overline{X}^2] = \frac{n-1}{n} V[X]$.
- $\overline{X^2} - \overline{X}^2 \xrightarrow{p} V[X]$.

Proof: Let $Z_i = X_i^2$, $\forall i \in \{1, 2, \dots, n\}$. It follows from a generalization of Theorem 1.3.17 (*Implications of Independence, Part I*) that $Z_1, Z_2, \dots, Z_n$ are i.i.d., so by Theorem 3.2.3 (*The Expected Value of the Sample Mean is the Population Mean*),

$$E[\overline{X^2}] = E[\overline{Z}] = E[Z] = E[X^2].$$

Now consider $\overline{X}^2$. Note that $E[\overline{X}^2] \neq E[X]^2$. Rather, since $V[\overline{X}] = E[\overline{X}^2] - E[\overline{X}]^2$,

$$E[\overline{X^2}] = E[\overline{X}]^2 + V[\overline{X}] = E[X]^2 + \frac{V[X]}{n},$$

for analogous population parameters (Goldberger 1968, Manski 1988). See Section 3.3 (*The Plug-In Principle*).

where the second equality follows from Theorem 3.2.3 and Theorem 3.2.4 (*Sampling Variance of the Sample Mean*). Thus,

$$\begin{aligned}
 E[\overline{X^2} - \overline{X}^2] &= E[\overline{X^2}] - E[\overline{X}^2] \\
 &= E[X^2] - \left(E[X]^2 + \frac{V[X]}{n} \right) \\
 &= \left(E[X^2] - E[X]^2 \right) - \frac{V[X]}{n} \\
 &= V[X] - \frac{V[X]}{n} \\
 &= \left(1 - \frac{1}{n} \right) V[X] \\
 &= \frac{n-1}{n} V[X].
 \end{aligned}$$

Now, by the WLLN, $\overline{X^2} \xrightarrow{P} E[X^2]$ and $\overline{X} \xrightarrow{P} E[X]$. Let $g(u, v) = u - v^2$. Then, by the CMT,

$$\overline{X^2} - \overline{X}^2 = g(\overline{X^2}, \overline{X}) \xrightarrow{P} g(E[X^2], E[X]) = E[X^2] - E[X]^2 = V[X]. \quad \square$$

Theorem 3.2.19 says that the plug-in sample variance is not unbiased but nevertheless converges in probability to the true variance under a mild regularity condition. The size of the bias decreases as n increases (since $\frac{n-1}{n} \rightarrow 1$ as $n \rightarrow \infty$), so in large samples, it becomes negligible. In this setting, it is also easy to correct for this bias. The *unbiased sample variance* (often referred to simply as the *sample variance*) incorporates this correction.

Definition 3.2.20. *Unbiased Sample Variance*

For i.i.d. random variables $X_1, X_2, \dots, X_n$, the *unbiased sample variance* is

$$\hat{V}[X] = \frac{n}{n-1} (\overline{X^2} - \overline{X}^2).$$

In general, just as we used $\hat{\theta}$ to denote an estimator of θ , we add a “hat” ($\hat{}$) to an expression to denote an estimator of that quantity. The following theorem states that the unbiased sample variance is indeed unbiased, and also consistent.

Theorem 3.2.21. *Properties of the Unbiased Sample Variance*

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite variance $V[X] > 0$,

- $E[\hat{V}[X]] = V[X]$.
- $\hat{V}[X] \xrightarrow{p} V[X]$.

Proof: By linearity of expectations and Theorem 3.2.19,

$$\begin{aligned} E[\hat{V}[X]] &= E\left[\frac{n}{n-1}(\overline{X^2} - \bar{X}^2)\right] \\ &= \frac{n}{n-1}E[\overline{X^2} - \bar{X}^2] \\ &= \frac{n}{n-1}\left(\frac{n-1}{n}V[X]\right) \\ &= V[X]. \end{aligned}$$

Now, by Theorem 3.2.19, $\overline{X^2} - \bar{X}^2 \xrightarrow{p} V[X]$, and treating $\frac{n}{n-1}$ as a sequence of degenerate random variables, $\frac{n}{n-1} \xrightarrow{p} 1$. Thus, by the CMT,

$$\hat{V}[X] = \frac{n}{n-1}(\overline{X^2} - \bar{X}^2) \xrightarrow{p} 1 \cdot V[X] = V[X]. \quad \square$$

As we will see, having a consistent estimator of $V[X]$ will allow us to quantify the uncertainty around the estimate of $E[X]$ given by the sample mean.

3.2.4 The Central Limit Theorem for Sample Means

We now present what is often considered the most profound and important theorem in statistics: the *Central Limit Theorem*. The Central Limit Theorem, loosely speaking, implies that the average of many i.i.d. random variables will follow a normal distribution. However, before we can state this theorem rigorously, we require a couple of definitions.

Definition 3.2.22. *Convergence in Distribution*

Let $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be a sequence of random variables with CDFs $(F_{(1)}, F_{(2)}, F_{(3)}, \dots)$, and let T be a random variable with CDF F . Then $T_{(n)}$ converges in distribution to T if, $\forall t \in \mathbb{R}$ at which F is continuous,

$$\lim_{n \rightarrow \infty} F_{(n)}(t) = F(t).$$

We write $T_{(n)} \xrightarrow{d} T$ to denote that $T_{(n)}$ converges in distribution to T .

The distribution of the random variable T to which a sequence of random variables $T_{(n)}$ converges in distribution is referred to as the *limit distribution* (or *asymptotic distribution*) of $T_{(n)}$. (This notion will be important in Section 3.2.5.) As with convergence in probability, this definition can apply to any sequence of random variables, but we use $T_{(n)}$ here because we will mainly be discussing convergence in distribution of sample statistics. We must also define the *standardized sample mean*.

Definition 3.2.23. Standardized Sample Mean

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite $E[X] = \mu$ and finite $V[X] = \sigma^2 > 0$, the *standardized sample mean* is

$$Z = \frac{(\bar{X} - E[\bar{X}])}{\sigma[\bar{X}]} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}.$$

It can be shown that $E[Z] = 0$ and $V[Z] = \sigma[Z] = 1$. These convenient properties are the reason why Z is called the standardized sample mean.

We can now state the Central Limit Theorem for sample means.

Theorem 3.2.24. Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with (finite) $E[X] = \mu$ and finite $V[X] = \sigma^2 > 0$, and let Z be the standardized sample mean. Then

$$Z \xrightarrow{d} N(0, 1),$$

or equivalently,

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2).$$

We omit the proof of this theorem.¹⁹ The Central Limit Theorem says that, if n is large, the sampling distribution of the sample mean will tend to be approximately normal even when the population under consideration is not distributed normally. This is useful because it will allow us to rigorously quantify our uncertainty when n is large, as we will see in Section 3.4 (*Inference*).

The equivalence of the two statements of the Central Limit Theorem follows from *Slutsky's Theorem*, which will be useful in our technical derivations going forward.

¹⁹ Full proofs of the Central Limit Theorem invariably require fairly advanced mathematics, though the proof of a special case, the De Moivre–Laplace Theorem, is relatively accessible and primarily relies on Sterling's approximation. See, e.g., Feller (1968, Sections II.9, VII.3).

Theorem 3.2.25. Slutsky's Theorem

Let $(S_{(1)}, S_{(2)}, S_{(3)}, \dots)$ and $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be sequences of random variables. Let T be a random variable and $c \in \mathbb{R}$. If $S_{(n)} \xrightarrow{D} c$ and $T_{(n)} \xrightarrow{D} T$, then

- $S_{(n)} + T_{(n)} \xrightarrow{D} c + T$.
- $S_{(n)} T_{(n)} \xrightarrow{D} cT$.
- $\frac{T_{(n)}}{S_{(n)}} \xrightarrow{D} \frac{T}{c}$, provided that $c \neq 0$.

We omit the proof of this theorem, but it can be derived from a variant of the Continuous Mapping Theorem (Theorem 3.2.7). We will often invoke this theorem for the special case in which $S_{(n)}$ is a degenerate random variable, either depending solely on n (for example, $\frac{n}{n-1} \xrightarrow{D} 1$) or just a constant (for example, $c \xrightarrow{D} c$). For example, the equivalence of the two statements of the Central Limit Theorem is shown as follows:

$$Z \xrightarrow{D} Z', \text{ where } Z' \sim N(0, 1).$$

By Slutsky's Theorem,

$$\sigma Z \xrightarrow{D} \sigma Z',$$

and

$$\sigma Z = \sigma \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \sqrt{n}(\bar{X} - \mu),$$

so

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} \sigma Z',$$

and by Theorem 2.1.21,

$$\sigma Z' \sim N(0, \sigma^2),$$

so $\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2)$. These steps can also be reversed, so the two statements are in fact equivalent.

Figure 3.2.2 illustrates the Central Limit Theorem in action. As the sample size increases, the distribution of the sample mean becomes both bell-shaped and increasingly tightly centered on the population mean. This holds regardless of the particular underlying distribution of X , as long as it has finite variance. However, sample means drawn from some distributions will more quickly approximate a normal distribution than others.

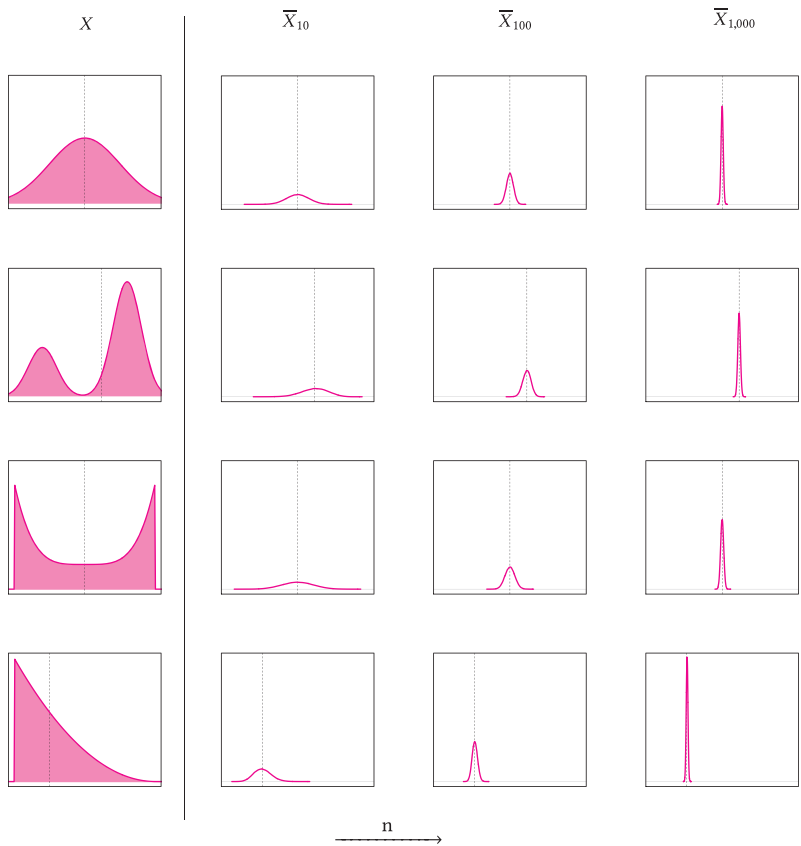


FIGURE 3.2.2. Simulated Sampling Distributions of the Sample Mean

3.2.5 Asymptotic Estimation Theory

We noted above that when a sample statistic $T_{(n)}$ converges in probability to a random variable T , the distribution of T is known as the limit distribution (or asymptotic distribution) of $T_{(n)}$. When we discuss the “asymptotic” properties of an estimator, we are referring to the features of its limit distribution (or some appropriate rescaling thereof).²⁰ We now define some such properties that may be desirable for estimators to have. We begin with a simple case: the asymptotic variant of the unbiasedness property from Section 3.2.2).

²⁰ There is not a consensus about nomenclature here; some authors instead refer to these properties as “limit” properties or simply as features of the limit distribution.

Definition 3.2.26. Asymptotic Unbiasedness

An estimator $\hat{\theta}$ is *asymptotically unbiased* for θ if $\hat{\theta} \xrightarrow{d} T$, where $E[T] = \theta$.

In other words, an estimator $\hat{\theta}$ is asymptotically unbiased if the expectation of its limit distribution is θ . This can introduce some small oddities. To see this, consider the following example.

Example 3.2.27. Asymptotic Unbiasedness Does Not Imply Unbiasedness Asymptotically

Suppose that $\forall n \in \mathbb{N}$, $\hat{\theta}_{(n)}$ has PMF

$$f_{(n)}(t) = \begin{cases} \frac{1}{n} & : t = \theta + n \\ \frac{n-1}{n} & : t = \theta. \end{cases}$$

Then, for any n ,

$$E[\hat{\theta}_{(n)}] = \frac{1}{n}(\theta + n) + \left(\frac{n-1}{n}\right)\theta = \frac{\theta}{n} + \frac{n}{n} + \theta \frac{n}{n} - \frac{\theta}{n} = \theta + 1.$$

So, regardless of n , $E[\hat{\theta}_{(n)}] = \theta + 1$. Thus, the estimator is never unbiased; it always has a bias of 1, even as n grows arbitrarily large:

$$\lim_{n \rightarrow \infty} (E[\hat{\theta}_{(n)}] - \theta) = 1.$$

Yet we can still say that $\hat{\theta}$ is asymptotically unbiased because $\hat{\theta}_{(n)} \xrightarrow{d} \theta$, and since θ is degenerate, $E[\theta] = \theta$.²¹ $\triangle$

This gives a sense of the “weirdness” that can occur when we are talking about asymptotics. Typically, it will make sense to employ a notion of asymptotics that focuses on the limit distribution, as we can safely ignore extreme events that become infinitesimally rare as n grows large.

Another asymptotic property that will be important going forward is asymptotic normality. We define this notion for a rescaling of an estimator, as we will only be concerned with this property as it applies to consistent estimators, which—since they converge in probability to a constant—have degenerate (unscaled) limit distributions.

Definition 3.2.28. Asymptotic Normality

An estimator $\hat{\theta}$ is *asymptotically normal* if $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \phi^2)$, for some finite $\phi > 0$.

²¹ Note that $\hat{\theta}$ is consistent, as $\hat{\theta}_{(n)} \xrightarrow{d} \theta$ implies $\hat{\theta}_{(n)} \xrightarrow{p} \theta$.

Asymptotic normality can thus be stated as follows: “The (rescaled) limit distribution of $\hat{\theta}$ is normal.” (Note that, due to the rescaling, our definition of asymptotic normality implies consistency.) Similarly, the Central Limit Theorem can be stated as follows: “The sample mean is asymptotically normal.” The notion of the limit distribution will also allow us to define an appropriate asymptotic analog to the standard error.

Definition 3.2.29. Asymptotic Standard Error

For an estimator $\hat{\theta}$ such that $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} T$, the *asymptotic standard error* of $\hat{\theta}$ is $\sqrt{V[T]}$.

The asymptotic standard error is simply the standard deviation of the limit distribution. We can also define the asymptotic analog to the mean squared error. Again, we define this notion in terms of a rescaling of the estimator, since the limit distribution of any consistent estimator is the degenerate random variable θ and thus has zero MSE.²²

Definition 3.2.30. Asymptotic MSE

For an estimator $\hat{\theta}$ such that $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} T$, the *asymptotic MSE* of $\hat{\theta}$ is $E[T^2]$.

This definition of asymptotic MSE allows us to compare the asymptotic efficiency of estimators, analogously to our definition of relative efficiency.

Definition 3.2.31. Asymptotic Relative Efficiency

Let $\hat{\theta}_A$ and $\hat{\theta}_B$ be estimators of θ . Then $\hat{\theta}_A$ is *asymptotically more efficient* than $\hat{\theta}_B$ if it has a lower asymptotic MSE.

This definition illustrates how the notions of efficiency generalize to limit distributions. We conclude this section by discussing a concept that will be essential for quantifying the uncertainty of estimates: consistency of a sampling variance estimator and consistency of a standard error estimator.

Definition 3.2.32. Consistency of Sampling Variance and Standard Error Estimators

For an estimator $\hat{\theta}$ such that $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} T$, a sampling variance estimator $\hat{V}[\hat{\theta}]$ is *consistent* if

$$n\hat{V}[\theta] \xrightarrow{p} V[T]$$

²² As with bias, however, consistency does not imply that $\lim_{n \rightarrow \infty} E[(\hat{\theta}_{(n)} - \theta)^2] = 0$.

and a standard error estimator $\sqrt{\hat{V}[\hat{\theta}]}$ is *consistent* if

$$\sqrt{n}\sqrt{\hat{V}[\theta]} \xrightarrow{p} \sqrt{V[T]}.^{23}$$

Note that, due to the CMT, consistency of the sampling variance estimator implies consistency of the corresponding standard error estimator, and vice versa. Consistency of these estimators, wedded to asymptotic normality, will be key to our discussion in Section 3.4 (*Inference*).

3.2.6 Estimating Standard Errors of Sample Means

So far, we have discussed the population variance $V[X]$, and an estimator thereof, namely, the unbiased sample variance, $\hat{V}[X]$. Additionally, in Section 3.2.1 (*Sample Means*), we showed that the sampling variance of the sample mean is $V[\bar{X}] = \frac{V[X]}{n}$. (Do not confuse these three quantities.) We now want to propose an estimator of the sampling variance of the sample mean, $\hat{V}[\bar{X}]$. The natural choice is $\frac{\hat{V}[X]}{n}$.

Theorem 3.2.33. *Estimating the Sampling Variance of the Sample Mean*
For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite variance $V[X] > 0$, let $\hat{V}[\bar{X}] = \frac{\hat{V}[X]}{n}$. Then

- $E[\hat{V}[\bar{X}]] = V[\bar{X}]$.
- $n\hat{V}[\bar{X}] - nV[\bar{X}] = n\hat{V}[X] - V[X] \xrightarrow{p} 0$.

Proof:

$$E[\hat{V}[\bar{X}]] = E\left[\frac{\hat{V}[X]}{n}\right] = \frac{E[\hat{V}[X]]}{n} = \frac{V[X]}{n} = V[\bar{X}],$$

and by Theorem 3.2.21, $\hat{V}[X] \xrightarrow{p} V[X]$, so since we can rearrange

$$\hat{V}[\bar{X}] = \frac{\hat{V}[X]}{n} \text{ and } V[\bar{X}] = \frac{V[X]}{n}$$

²³ Note that it is equivalent to define these as $n\hat{V}[\theta] - V[T] \xrightarrow{p} 0$ and $\sqrt{n}\sqrt{\hat{V}[\theta]} - \sqrt{V[T]} \xrightarrow{p} 0$, respectively.

to get

$$n\hat{V}[\bar{X}] = \hat{V}[X] \text{ and } nV[\bar{X}] = V[X],$$

by substitution, $n\hat{V}[\bar{X}] - nV[\bar{X}] \xrightarrow{p} 0$. $\square$

The first property states that the sampling variance of the sample mean is unbiased, and the second is equivalent to stating that it is consistent (as the limit distribution of the sample mean has variance $V[X]$). Thus, it is possible to estimate the average of a random variable (the population mean) and, furthermore, to estimate the uncertainty of that estimate. (We will formally discuss quantifying the uncertainty of an estimate in Section 3.4, *Inference*.)

As with any variance, we can take the square root of the sampling variance of the sample mean to obtain the standard error, or the standard deviation of the sampling distribution of the sample mean.

Definition 3.2.34. *Standard Error of the Sample Mean*

For i.i.d. random variables $X_1, X_2, \dots, X_n$, the *standard error of the sample mean* is

$$\sigma[\bar{X}] = \sqrt{V[\bar{X}]}.$$

To estimate the standard error of the sample mean, we use $\hat{\sigma}[\bar{X}] = \sqrt{\hat{V}[\bar{X}]}$. When we see “standard error” in a paper, this is what the authors really mean: an estimate thereof. We do not know the true standard error.²⁴ This estimator is not unbiased,²⁵ but it is consistent.

Theorem 3.2.35. *Consistency of the Standard Error of the Sample Mean Estimator*

²⁴ Since an estimator of the standard error is itself a random variable, it has its own sampling distribution and thus its own standard error, so we could, in theory, estimate the standard error of the standard error of the sample mean, thereby quantifying the uncertainty of our estimate of the uncertainty of our estimate of the population mean. However, since standard error estimators converge “quickly” to the true standard error, this is not a concern with large n .

²⁵ This is due to Jensen’s Inequality: for a random variable X and convex function g of X , $E[g(X)] \geq g(E[X])$. Corollary: for a random variable X and concave function g of X , $E[g(X)] \leq g(E[X])$. These inequalities are strict when g is strictly convex/concave and X is not degenerate. The square root function is strictly concave, so whenever $V[\hat{V}[\bar{X}]] > 0$,

$$E[\hat{\sigma}[\bar{X}]] = E\left[\sqrt{\hat{V}[\bar{X}]}\right] < \sqrt{E[\hat{V}[\bar{X}]]} = \sqrt{V[\bar{X}]} = \sigma[\bar{X}].$$

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with finite variance $V[X] > 0$,

$$\sqrt{n}\hat{\sigma}[\bar{X}] - \sqrt{V[X]} = \sqrt{n}\hat{\sigma}[\bar{X}] - \sqrt{n}\sigma[\bar{X}] \xrightarrow{p} 0.$$

Proof: Let $g(u) = \sqrt{u}$. By Theorem 3.2.33, $n\hat{V}[\bar{X}] - nV[\bar{X}] \xrightarrow{p} 0$. So, since g is continuous, by the CMT,

$$\sqrt{n}\hat{\sigma}[\bar{X}] = \sqrt{\hat{V}[\bar{X}]} = g(\hat{V}[\bar{X}]) \xrightarrow{p} g(V[\bar{X}]) = \sqrt{V[\bar{X}]} = \sqrt{n}\sigma[\bar{X}]. \quad \square$$

The ideas used in establishing the properties of the sample mean can take us very far. In the following section, we formalize a notion of estimation that will allow us to straightforwardly estimate well-defined population quantities with sample analogs.

3.3 THE PLUG-IN PRINCIPLE²⁶

The estimation philosophy used in this chapter has been informally based on the *plug-in principle*: write down the feature of the population that we are interested in, and then use the sample analog to estimate it. For example, if we want to estimate the expected value, use the sample mean. If we want to estimate the population variance, use the sample variance. It is possible to formalize and generalize this idea, so that it encompasses a general framework for estimation. In general, the CDF of a random variable tells us everything about the behavior of the random variable. The CDF of X is

$$F(x) = \Pr[X \leq x] = E[I(X \leq x)].$$

Accordingly, there is a sample analog to the CDF, the *empirical CDF*.

Definition 3.3.1. Empirical CDF

For i.i.d. random variables $X_1, X_2, \dots, X_n$, the *empirical CDF* $\hat{F}$ of X is

$$\hat{F}(x) = \overline{I(X \leq x)}, \forall x \in \mathbb{R}.^{27}$$

Just as the CDF is a function that fully describes the population, the empirical CDF is a function that fully describes the sample. Note that the empirical CDF is always a nondecreasing step function, like the CDF of a discrete random variable. Figure 3.3.1 illustrates that this is the case even when one's

²⁶ This section is heavily inspired by the treatments in Wasserman (2004), Wasserman (2006), and Wasserman (2012).

²⁷ This definition, and everything that follows, can easily be generalized to the multivariate case. For example, in the bivariate case, for i.i.d. random vectors $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$, the *joint* empirical CDF of (X, Y) is $\hat{F}(x, y) = \overline{I(X \leq x, Y \leq y)}$. Everything else proceeds analogously.

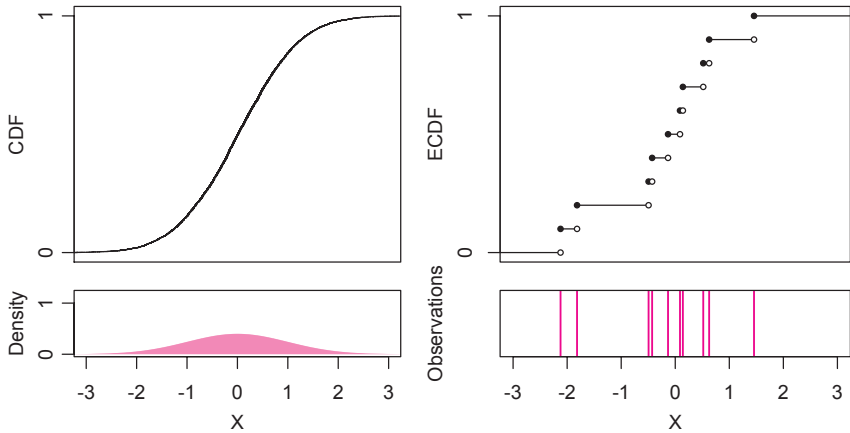


FIGURE 3.3.1. CDF of a Continuous Random Variable, and Illustrative Empirical CDF

sample is drawn from a continuous random variable (for which the CDF is continuous by definition, see Section 1.2.4, *Continuous Random Variables*). There are two (equivalent) ways of thinking about how $\hat{F}(x)$ is computed:

- Replace the expected value in the definition of the CDF with a sample mean.²⁸
- Count the number of observations i satisfying $X_i \leq x$ and divide by n .

We know, because of the properties of the sample mean, that, $\forall x \in \mathbb{R}$, $\hat{F}(x)$ will be unbiased and consistent for $F(x)$ (see Theorem 3.2.9, *Estimating the CDF*).²⁹

Before we proceed, we take a moment to establish some necessary notation. Let F be the CDF of a random variable X , and let f be the corresponding probability mass function (PMF) or probability density function (PDF). Let g be any function of X . Then let

$$\int_{-\infty}^{\infty} g(x) dF(x) = \begin{cases} \sum_x g(x) f(x) & : F \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & : F \text{ is continuous.} \end{cases}^{30}$$

²⁸ Recall from Definition 3.2.2 (*Sample Mean*) that $\overline{I(X \leq x)}$ denotes the sample mean of $I(X \leq x)$.

²⁹ In fact, an even stronger claim holds. The Glivenko–Cantelli Theorem:

$$\sup_x |\hat{F}(x) - F(x)| \xrightarrow{a.s.} 0,$$

where the notation $\xrightarrow{a.s.}$ denotes *almost sure convergence*, which is a notion of convergence stronger than convergence in probability. (We will not discuss this type of convergence in detail.) In words, the Glivenko–Cantelli Theorem states that the biggest difference between the CDF and empirical CDF converges to zero. In fact, it converges to zero “quickly.” See the Dvoretzky–Kiefer–Wolfowitz inequality.

We can now introduce the concept of a *statistical functional*. Simply put, a statistical functional is an operator on a random variable that is represented by a function of the CDF. It is simply a way to represent a feature of a random variable.

Definition 3.3.2. Statistical Functional

A *statistical functional* is a function $T: \mathcal{F} \rightarrow \mathbb{R}$, where $\mathcal{F}$ is the set of all possible CDFs.

For example, for the random variable X with CDF F , we can write the expected value of X as

$$E[X] = T_E(F) = \int_{-\infty}^{\infty} x dF(x).$$

Likewise, we can write the variance of X as

$$\begin{aligned} V[X] &= T_V(F) \\ &= \int_{-\infty}^{\infty} (x - E[X])^2 dF(x) \\ &= \int_{-\infty}^{\infty} x^2 dF(x) - E[X]^2 \\ &= \int_{-\infty}^{\infty} x^2 dF(x) + \left(\int_{-\infty}^{\infty} x dF(x) \right)^2. \end{aligned}$$

A remarkably large class of estimands can be written as $\theta = T(F)$, and the inferential targets that we will set in this book all have representations as statistical functionals. With θ as our target, we can define the *plug-in estimator* of θ as follows:

Definition 3.3.3. Plug-In Estimator

For i.i.d. random variables $X_1, X_2, \dots, X_n$ with common CDF F , the *plug-in estimator* of $\theta = T(F)$ is

$$\hat{\theta} = T(\hat{F}).$$

To estimate $\theta = T(F)$, apply the same functional to the *empirical CDF*. We provide two examples showing how we could derive plug-in estimators for the population mean and variances.

³⁰ These are special cases of Riemann–Stieltjes integration, which is a generalization of ordinary (Riemann) integration to encompass discrete, continuous, and mixed-type distributions. This in turn can be generalized further with Lebesgue integration.

Example 3.3.4. *Sample Mean as a Plug-In Estimator*

Since the expected value of X is $E[X] = T_E(F) = \int_{-\infty}^{\infty} x dF(x)$, the plug-in estimator of $E[X]$ is

$$\begin{aligned}
 \hat{E}[X] &= T_E(\hat{F}) \\
 &= \int_{-\infty}^{\infty} x d\hat{F}(x) \\
 &= \sum_x x \hat{f}(x) \\
 &= \sum_x x \frac{(\text{number of observations } i \text{ with outcome } X_i = x)}{n} \\
 &= \frac{1}{n} \sum_x x (\text{number of observations } i \text{ with outcome } X_i = x) \\
 &= \frac{1}{n} \sum_{i=1}^n x = \bar{X},
 \end{aligned}$$

where $\hat{f}$ is the empirical PMF. Thus, the sample mean is the plug-in estimator of the expected value. $\triangle$

The analogous result for the population variance follows directly.

Example 3.3.5. *Plug-In Sample Variance as a Plug-In Estimator*

Since the variance of X is

$$V[X] = T_V(F) = \int_{-\infty}^{\infty} x^2 dF(x) + \left(\int_{-\infty}^{\infty} x dF(x) \right)^2,$$

the plug-in estimator of $V[X]$ is

$$\hat{V}[X] = T_V(\hat{F}) = \int_{-\infty}^{\infty} x^2 d\hat{F}(x) - \left(\int_{-\infty}^{\infty} x d\hat{F}(x) \right)^2 = \bar{X}^2 - \bar{X}^2.$$

Thus, the plug-in sample variance is, in fact, the plug-in estimator of the variance. $\triangle$

The idea of a plug-in estimator is quite profound. We want to know a feature θ of the CDF. We express θ as a function T of the CDF. We then observe the sample analog of the CDF, namely, the empirical CDF. We know that, as n grows large, the empirical CDF tends to more and more closely approximate the CDF. We therefore estimate θ by applying the function T to the empirical CDF. Then, as n grows large, the estimate θ tends to more and more closely approximate θ . Typically, we do not need to go to the extreme of writing everything in terms of the CDF or empirical CDF, as we can derive most plug-in estimators by substituting sample quantities for population quantities (for instance, means for expectations). Nonetheless, when we talk about a plug-in

estimator, we are fundamentally talking about an estimator that substitutes (“plugs in”) the empirical CDF for the CDF.

3.3.1 The Usual Plug-In Regularity Conditions

Regularity conditions sufficient for plug-in estimators to have good asymptotic behavior (such as consistency and asymptotic normality) usually entail restrictions on the manner in which small changes in the empirical CDF can correspond to changes in the estimate. In particular, we assume that T is suitably *well behaved* at F ; if we were to perturb the true population distribution F just a little bit, we would not expect the value that the functional takes on to change dramatically. This is perhaps intuitive: if the functional were prone to take on drastically different values for CDFs that are very close to the population distribution, we would not expect an estimator based on the empirical CDF to converge. We provide an informal statement of a key result (Wasserman 2006, Theorem 2.27), namely that mild regularity conditions are sufficient to guarantee that a plug-in estimator is asymptotically normal.

Theorem 3.3.6. *Asymptotic Normality of Plug-In Estimators*

If T is well behaved³¹ at F and $\hat{\theta} = T(\hat{F})$, then $\hat{\theta}$ is asymptotically normal.

We omit the proof, which requires empirical process theory. The conditions for T to be well behaved at F are often implied by other assumptions; for instance, our assumption about finite variance was sufficient to imply that the population mean is well behaved. We will refer to these assumptions, as they apply to the functional under consideration, as the *usual plug-in regularity conditions*. The usual plug-in regularity conditions can be satisfied by assuming (and even verifying, in some cases) that the functional of interest will be well behaved in the neighborhoods of plausible population distributions.

While it can fail, the plug-in principle remains a good starting point in considering estimation strategies. In the next section, we will describe another way of estimating the distribution of a random variable, which will in turn yield natural plug-in estimators.

³¹ We say T is well behaved if it is Hadamard differentiable at F with respect to the Kolmogorov distance. Formally, the Kolmogorov distance between two distributions F and G is given by the value of the largest divergence between their two CDFs, $\sup_x |F(x) - G(x)|$. Attentive readers may recall that the Glivenko–Cantelli Theorem in Footnote 29 guarantees almost sure convergence of the Kolmogorov distance between $\hat{F}$ and F to zero, providing further intuition for Theorem 3.3.6. Although we do not provide a formal definition, Wasserman (2006, Ch. 2) provides a lucid discussion of the assumption of Hadamard differentiability.

3.3.2 Kernel Estimation

The procedure illustrated above is easily applicable for discrete random variables. But what if we are dealing with a continuous random variable, and we would like to write our target in terms of the PDF? If we are willing to impose some strong assumptions about the distribution of the random variable, it becomes fairly easy to estimate its PDF (see Chapter 5). But what if we do not want to make such strong assumptions? In that case, we can still estimate the PDF using *kernel density estimation*.

The key idea behind kernel density estimation is to “smooth” the data so as to get estimates of the PDF everywhere. We can estimate the value of the PDF at a given point x by calculating the proportion of the observations that have X_i near x .

Definition 3.3.7. Kernel Density Estimator

Let $X_1, X_2, \dots, X_n$ be i.i.d. continuous random variables with common PDF f . Let $K: \mathbb{R} \rightarrow \mathbb{R}$ be an even function³² satisfying $\int_{-\infty}^{\infty} K(x)dx = 1$, and let $K_b(x) = \frac{1}{b}K(\frac{x}{b})$, $\forall x \in \mathbb{R}$ and $b > 0$. Then a *kernel density estimator* of $f(x)$ is

$$\hat{f}_K(x) = \frac{1}{n} \sum_{i=1}^n K_b(x - X_i), \forall x \in \mathbb{R}.$$

The function K is called the *kernel* and the scaling parameter b is called the *bandwidth*.

The kernel and bandwidth tell us how much to weight each observation X_i depending on its distance from the point of interest x . The farther an observation is from the point of interest, the less it counts. This is easiest to understand in the univariate case, though everything carries through to multiple variables as long as “distance” is defined accordingly. (Regular Euclidean distance is the usual choice.) The kernel tells us the shape of the weighting, and the bandwidth tells us how far out we should go, that is, the width or scale of the kernel. There are many common kernels (see Figure 3.3.2). Then computing estimates of the densities at every point follows directly: we add up all the observations, weighted by a metric of how much we want each one to count. The closer it is to the point of interest, the more it counts.

In order for kernel density estimates to be consistent, we will need the bandwidth b to tend to zero as n tends to infinity. Then, as long as the PDF really is continuous, kernel density estimation will be consistent.³³ This follows

³² That is, a function whose graph is symmetric about the y-axis, so that $\forall x \in \mathbb{R}, f(x) = f(-x)$.

³³ We also want to make sure that $b \rightarrow 0$ at an appropriate rate, so that we are never over or undersmoothing. Silverman (1986)’s “rule of thumb” for kernel density estimation with a

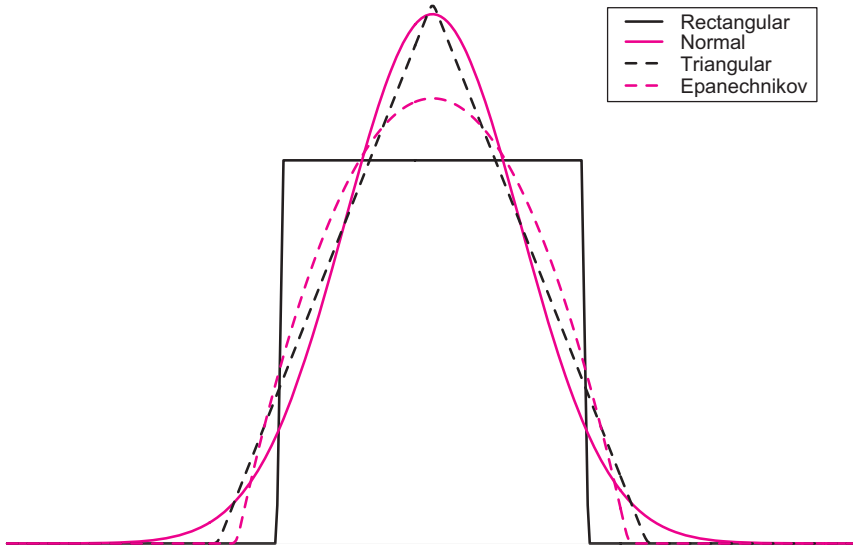


FIGURE 3.3.2. Visualization of Different Kernels

intuitively from the definition of the PDF:

$$f(x) = \frac{dF(x)}{dx} = \lim_{h \rightarrow 0^+} \frac{F(x+h) - F(x-h)}{2h}.$$

The following example shows how, for continuous X , the kernel density estimate (with a rectangular kernel) will be consistent for the true PDF. The proof itself leans on the plug-in principle.

Theorem 3.3.8. *Consistency of the Kernel Density Estimator (Rectangular Kernel)*

For i.i.d. continuous random variables $X_1, X_2, \dots, X_n, \forall x \in \mathbb{R}$, if $K(x) = \frac{1}{2}I(-1 \leq x \leq 1)$ (that is, the kernel is rectangular), and if $h \rightarrow 0$ and $nh \rightarrow \infty$ as $n \rightarrow \infty$, then $\hat{f}_K(x) \xrightarrow{P} f(x)$.

normal (Gaussian) kernel is to let

$$h = \frac{0.9 \min\left(\hat{\sigma}[X], \frac{IQR}{1.34}\right)}{n^{\frac{1}{5}}},$$

where $\hat{\sigma}[X] = \sqrt{\hat{V}[X]}$ and IQR is the (sample) interquartile range. Under this rule (and mild regularity conditions), the range of observations that we count toward a point's density is going to shrink at a rate on the order of $n^{-\frac{1}{5}}$. That is pretty slow, but eventually, with large enough n , we will only count observations that are very close.

Proof sketch: Our proof sketch relies on the plug-in principle. For any $h > 0$,

$$K_b(x) = \frac{1}{2b} \mathbf{I}\left(-1 \leq \frac{x}{b} \leq 1\right) = \frac{\mathbf{I}(-b \leq x \leq b)}{2b}.$$

Thus,

$$\begin{aligned} \hat{f}_K(x) &= \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{I}(-b \leq X_i - x \leq b)}{2b} \\ &= \frac{\frac{1}{n} \sum_{i=1}^n \mathbf{I}(x - b \leq X_i \leq x + b)}{2b} \\ &= \frac{\hat{F}(x + b) - \hat{F}(x - b)}{2b} \\ &\xrightarrow{p} \frac{F(x + b) - F(x - b)}{2b}, \end{aligned}$$

which follows from the definition of the empirical CDF and its consistency for F . Then, we know that the definition of the PDF is simply a limit of the final expression as $h \rightarrow 0$:

$$\lim_{h \rightarrow 0} \frac{F(x + b) - F(x - b)}{2b} = \frac{dF(x)}{dx} = f(x).$$

The condition that $nh \rightarrow \infty$ is sufficient to guarantee that this limit carries through in estimation. $\square$

Now, once we have an empirical estimate of the PDF, we can compute any quantity that we want from the estimate. We can integrate our estimate of the PDF to obtain an estimate of the full CDF, and thus we can apply any statistical functional to that estimate. This provides a basis for another type of plug-in estimation: kernel plug-in estimation.

Definition 3.3.9. Kernel Plug-In Estimator

The *kernel plug-in estimator* of $\theta = T(F)$ is

$$\hat{\theta}_K = T(\hat{F}_K),$$

where $\hat{F}_K(x) = \int_{-\infty}^x \hat{f}(u) du$.

The only difference between plug-in estimation and kernel plug-in estimation is the way that we estimate the CDF. In plug-in estimation, we use the empirical CDF $\hat{F}$. In kernel plug-in estimation, we use the smooth estimate of the CDF implied by the kernel density estimator.

The main benefit of the kernel-based approach comes in the multivariate case, since by obtaining a continuous estimate of the joint PDF, we can directly obtain point predictions even when we need to condition on a continuous variable. For example, if we used kernel density estimation to estimate $f_{Y|X}(y|\mathbf{x})$ (that is, use kernel estimation to estimate $f(y, \mathbf{x})$ and $f_X(\mathbf{x})$, and then apply Definition 1.3.14, *Conditional PDF*), then we could use our estimate $\hat{f}_{Y|X}(y|\mathbf{x})$ to estimate the conditional expectation function (CEF) at any given point $\mathbf{x}$:

$$\hat{E}[Y|X = \mathbf{x}] = \int_{-\infty}^{\infty} y \hat{f}_{Y|X}(y|\mathbf{x}) dy.$$

As long as the density estimator is consistent (in an appropriately uniform sense), all of the associated plug-in estimators will be consistent. This holds for any reasonable feature of the random variable that we might be interested in (such as conditional moments).

3.4 INFERENCE

We have shown how to estimate features of a random variable or population from i.i.d. samples. But how do we know how certain we can be that these estimates closely approximate reality? Can we precisely quantify the degree of uncertainty surrounding our estimates? These questions are the subject of statistical *inference*. In this section, among other things, we shall see the importance of our discussion of the standard error and the Central Limit Theorem in Section 3.2 (*Estimation*).

3.4.1 Confidence Intervals

A *confidence interval* for θ , loosely speaking, is an estimated *interval* that covers the true value of θ with at least a given probability. For example, if we were to take an i.i.d. sample of size n of a random variable X and compute a (valid) 95% confidence interval for the population mean $E[X]$, then with 95% probability, this interval will include $E[X]$. We now provide a formal definition of a confidence interval.

Definition 3.4.1. Valid Confidence Interval

A *valid confidence interval* for θ with *coverage* $(1 - \alpha)$ is a random interval $CI_{(1-\alpha)}(\theta)$,³⁴ such that $\Pr[\theta \in CI_{(1-\alpha)}(\theta)] \geq 1 - \alpha$.

For any given $\alpha \in (0, 1)$, we say that the *confidence level* is $100(1 - \alpha)\%$ and that $CI_{1-\alpha}(\theta)$ is a $100(1 - \alpha)\%$ confidence interval for θ . Some confidence

³⁴ A *random interval* is simply an interval formed by a pair of random variables. Suppose we have two random variables C_1, C_2 , with $\Pr[C_1 < C_2] = 1$. Then $C = (C_1, C_2)$ is an (open) random interval.

intervals are only valid asymptotically; they only have coverage $(1 - \alpha)$ with very large n . By *asymptotically valid*,³⁵ we mean that

$$\lim_{n \rightarrow \infty} \Pr[\theta \in CI_{1-\alpha}(\theta)] \geq 1 - \alpha.$$

The types of confidence intervals we will consider in this book will only be asymptotically valid. The following theorem gives a remarkably general method for obtaining asymptotically valid confidence intervals for asymptotically normal estimators.

Theorem 3.4.2. *Normal Approximation–Based Confidence Interval*

Let $\hat{\theta}$ be an asymptotically normal estimator of θ , $\hat{V}[\hat{\theta}]$ be a consistent estimator of the sampling variance, and $\alpha \in (0, 1)$. Then an asymptotically valid normal approximation–based confidence interval for θ with coverage $(1 - \alpha)$ is given by

$$CI_{1-\alpha}(\theta) = \left(\hat{\theta} - z_{1-\frac{\alpha}{2}} \sqrt{\hat{V}[\hat{\theta}]}, \hat{\theta} + z_{1-\frac{\alpha}{2}} \sqrt{\hat{V}[\hat{\theta}]} \right),$$

where z_c denotes the c^{th} quantile of the standard normal distribution; that is, $\Phi(z_c) = c$, $\forall c \in (0, 1)$.

Proof: By asymptotic normality,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \phi^2).$$

Define the random variable

$$Z = \frac{\sqrt{n}(\hat{\theta} - \theta)}{\phi},$$

so that by Slutsky's Theorem and Theorem 2.1.21, $Z \xrightarrow{d} N(0, 1)$.

Recall that by the CMT and consistency of the sampling variance estimator,

$$\sqrt{n\hat{V}[\hat{\theta}]} \xrightarrow{p} \phi.$$

Now, we plug in the *estimated* standard error. Define the random variable

$$Z' = \frac{\hat{\theta} - \theta}{\sqrt{\hat{V}[\hat{\theta}]}} = \frac{\hat{\theta} - \theta}{\sqrt{\hat{V}[\hat{\theta}]}} \frac{\sqrt{n}\phi}{\sqrt{n}\phi} = \frac{\sqrt{n}(\hat{\theta} - \theta)}{\phi} \frac{\phi}{\sqrt{n}\sqrt{\hat{V}[\hat{\theta}]}} = Z \frac{\phi}{\sqrt{n\hat{V}[\hat{\theta}]}}.$$

³⁵ There is a technical distinction between *pointwise* and *uniform* asymptotic validity, the latter being a stronger notion. Here we consider pointwise asymptotic validity.

Then, by the CMT and consistency of the sampling variance estimator,

$$\frac{\phi}{\sqrt{n\hat{V}[\hat{\theta}]}} \xrightarrow{p} 1.$$

So, by Slutsky's Theorem, $Z' \xrightarrow{d} N(0,1)$. Now, take an arbitrary $\alpha \in (0,1)$. By symmetry of the normal distribution, $z_{\frac{\alpha}{2}} = -z_{1-\frac{\alpha}{2}}$. Since $Z' \xrightarrow{d} N(0,1)$, it follows that

$$\begin{aligned} \lim_{n \rightarrow \infty} \Pr\left[-z_{1-\frac{\alpha}{2}} \leq Z' \leq z_{1-\frac{\alpha}{2}}\right] &= \lim_{n \rightarrow \infty} \left(F_{Z'}\left(z_{1-\frac{\alpha}{2}}\right) - F_{Z'}\left(-z_{1-\frac{\alpha}{2}}\right)\right) \\ &= \lim_{n \rightarrow \infty} \left(F_{Z'}\left(z_{1-\frac{\alpha}{2}}\right) - F_{Z'}\left(z_{\frac{\alpha}{2}}\right)\right) \\ &= \lim_{n \rightarrow \infty} F_{Z'}\left(z_{1-\frac{\alpha}{2}}\right) - \lim_{n \rightarrow \infty} F_{Z'}\left(z_{\frac{\alpha}{2}}\right) \\ &= \Phi\left(z_{1-\frac{\alpha}{2}}\right) - \Phi\left(z_{\frac{\alpha}{2}}\right) \\ &= 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha, \end{aligned}$$

where $F_{Z'}$ denotes the CDF of Z' . Thus,

$$\begin{aligned} \lim_{n \rightarrow \infty} \Pr[\theta \in CI_{1-\alpha}(\theta)] &= \lim_{n \rightarrow \infty} \Pr\left[\hat{\theta} - z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]} \leq \theta \leq \hat{\theta} + z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]}\right] \\ &= \lim_{n \rightarrow \infty} \Pr\left[-z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]} \leq \theta - \hat{\theta} \leq z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]}\right] \\ &= \lim_{n \rightarrow \infty} \Pr\left[-z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]} < \hat{\theta} - \theta < z_{1-\frac{\alpha}{2}}\sqrt{\hat{V}[\hat{\theta}]}\right] \\ &= \lim_{n \rightarrow \infty} \Pr\left[-z_{1-\frac{\alpha}{2}} < \frac{\hat{\theta} - \theta}{\sqrt{\hat{V}[\hat{\theta}]}} < z_{1-\frac{\alpha}{2}}\right] \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \Pr \left[-z_{1-\frac{\alpha}{2}} < Z' < z_{1-\frac{\alpha}{2}} \right] \\
&= 1 - \alpha. \quad \square
\end{aligned}$$

Specific values of $z_{1-\frac{\alpha}{2}}$ can be obtained using a Z table, graphing calculator, or statistical software, but it is useful to know the most commonly used values:

- For a 90% confidence interval, $\alpha = 0.10$ and $z_{1-\frac{\alpha}{2}} \approx 1.64$.
- For a 95% confidence interval, $\alpha = 0.05$ and $z_{1-\frac{\alpha}{2}} \approx 1.96$.
- For a 99% confidence interval, $\alpha = 0.01$ and $z_{1-\frac{\alpha}{2}} \approx 2.58$.

To ground these ideas, we provide an example of how to form a confidence interval for the population mean.

Example 3.4.3. *Confidence Interval for the Population Mean*

Suppose we had n i.i.d. draws of a random variable X . Then an approximate 95% confidence interval for the population mean $\mu = E[X]$ would be

$$\begin{aligned}
CI_{.95}(\mu) &= \left(\bar{X} - 1.96 \sqrt{\frac{\hat{V}[X]}{n}}, \bar{X} + 1.96 \sqrt{\frac{\hat{V}[X]}{n}} \right) \\
&= \left(\bar{X} - 1.96 \sqrt{\hat{V}[\bar{X}]}, \bar{X} + 1.96 \sqrt{\hat{V}[\bar{X}]} \right) \\
&= \left(\bar{X} - 1.96 \hat{\sigma}[\bar{X}], \bar{X} + 1.96 \hat{\sigma}[\bar{X}] \right).
\end{aligned}$$

If n is large, then, with 95% probability, $CI_{.95}(\mu)$ will encompass the true value of μ . Δ

Recall, however, that these confidence intervals are only guaranteed to have proper coverage rates if n is large—this is the “asymptotically valid” qualification. This implies that, in small samples, 95% normal approximation–based confidence intervals may actually have, for example, 80% coverage. This is in part attributable to the fact that the standard error of the estimator must be estimated, and is not known—a problem that does not admit a general solution without additional assumptions.³⁶ This qualification will carry through to our next topic, hypothesis testing, and will be discussed more generally in Section 3.4.4 (*Micronumerosity*).

³⁶ Another reason for the failure in finite samples is that the sampling distribution of the estimator is only approximately normal. This approximation may be bad in finite samples. This is in part addressed in Section 3.4.3 (*The Bootstrap*), when we discuss the percentile bootstrap.

3.4.2 Hypothesis Testing

The foregoing discussion of confidence intervals naturally leads to the closely related idea of hypothesis testing. Suppose we wanted to test the *null hypothesis* that $\theta = \theta_0$. (The null hypothesis is simply the hypothesis that we are testing against.) That is, having observed our sample and computed an estimate $\hat{\theta}$, we could ask: if θ were actually equal to θ_0 , what would be the probability that we would have obtained a $\hat{\theta}$ at least as far off from θ_0 as we did? This probability is called a *p-value*.

Definition 3.4.4. *p-Value*³⁷

Let $\hat{\theta}$ be an estimator of θ and let $\hat{\theta}^*$ be the observed value of $\hat{\theta}$. Then

- a *lower one-tailed p-value* under the null hypothesis $\theta = \theta_0$ is

$$p = \Pr_{\theta_0}[\hat{\theta} \leq \hat{\theta}^*],$$

- an *upper one-tailed p-value* under the null hypothesis $\theta = \theta_0$ is

$$p = \Pr_{\theta_0}[\hat{\theta} \geq \hat{\theta}^*],$$

- a *two-tailed p-value* under the null hypothesis $\theta = \theta_0$ is

$$p = \Pr_{\theta_0}[|\hat{\theta} - \theta_0| \geq |\hat{\theta}^* - \theta_0|],$$

where $\Pr_{\theta_0}[\cdot]$ denotes the probability under the assumption that $\theta = \theta_0$.

Intuitively, a low *p-value* means “if the null hypothesis (that $\theta = \theta_0$) were true, we would infrequently encounter a result as extreme as the one that we saw. Therefore, if we *reject* the null hypothesis (that is, if we conclude that $\theta \neq \theta_0$) based solely on how extreme the result is, then that decision will be a mistake either infrequently (if $\theta = \theta_0$) or never (if $\theta \neq \theta_0$).” Thus, a low *p-value* offers probabilistic guarantees for those attempting to reject a null hypothesis. Note that a high *p-value* does not offer the same guarantees for those looking to *accept* a null hypothesis and is accordingly limited in its utility for decision making.

There is a general approach for obtaining asymptotically valid *Wald-type p-values* using the same principles as Theorem 3.4.2 (*Normal Approximation-Based Confidence Interval*). First, we will define a useful statistic, known as

³⁷ This is only one way of computing a *p-value*. It is possible to test hypotheses about θ using any variety of *test statistics*; we focus here on the case of using $\hat{\theta}$ as a test statistic.

the t -statistic. The t -statistic is a summary metric of how unusual the observed value of $\hat{\theta}$ would be under the null hypothesis that $\theta = \theta_0$.³⁸

Definition 3.4.5. t -Statistic

Let $\hat{\theta}$ be an asymptotically normal estimator of θ , $\hat{V}[\hat{\theta}]$ be a consistent estimator of the sampling variance, and $\hat{\theta}^*$ be the observed value of $\hat{\theta}$. Then the t -statistic is

$$t = \frac{\hat{\theta}^* - \theta_0}{\sqrt{\hat{V}[\hat{\theta}]}}.$$

The utility of the t -statistic immediately becomes evident in the following theorem, which gives a general method for obtaining asymptotically valid p -values using the same principles as Theorem 3.4.2 (*Normal Approximation-Based Confidence Interval*).

Theorem 3.4.6. Normal Approximation-Based p -Values

Let $\hat{\theta}$ be an asymptotically normal estimator of θ , $\hat{V}[\hat{\theta}]$ be a consistent estimator of the sampling variance, and $\hat{\theta}^*$ be the observed value of $\hat{\theta}$. Then

- an asymptotically valid lower one-tailed p -value under the null hypothesis that $\theta = \theta_0$ is given by

$$p = \Phi \left(\frac{\hat{\theta}^* - \theta_0}{\sqrt{\hat{V}[\hat{\theta}]}} \right) = \Phi(t).$$

- an asymptotically valid upper one-tailed p -value under the null hypothesis that $\theta = \theta_0$ is given by

$$p = 1 - \Phi \left(\frac{\hat{\theta}^* - \theta_0}{\sqrt{\hat{V}[\hat{\theta}]}} \right) = 1 - \Phi(t).$$

³⁸ The t -statistic is so named because it is known to exactly follow Student's t -distribution under the null hypothesis that $\theta = \theta_0$ for certain linear estimators under a finite-sample normality assumption. We avoid discussion of the t -distribution as, all else equal, tests using the t -distribution and using a normal approximation asymptotically agree with i.i.d. data.

- an asymptotically valid two-tailed p -value under the null hypothesis that $\theta = \theta_0$ is given by

$$p = 2 \left(1 - \Phi \left(\frac{|\hat{\theta}^* - \theta_0|}{\sqrt{\hat{V}[\hat{\theta}]}} \right) \right) = 2 \left(1 - \Phi(|t|) \right).$$

We omit the proof of this theorem, as well as a formal definition of asymptotic validity. Loosely speaking, however, asymptotic validity in the context of p -values implies that, under the null hypothesis that $\theta = \theta_0$, as $n \rightarrow \infty$, $p \xrightarrow{d} U(0, 1)$. (For example, if $\theta = \theta_0$, n is large and the p -value is asymptotically valid, then across repeated samples, we would only obtain $p \leq 0.05$ in 5% of samples.) As with Section 3.4.1 (*Confidence Intervals*), asymptotically valid p -values do not have guarantees for finite samples. (See Section 3.4.4.)

One direct implication of Theorem 3.4.6 is that an asymptotically valid p -value is given by a one-to-one function of t . For example, if $|t| = 1.96$, then an asymptotically valid two-tailed p -value is given by $p = 0.05$. Traditionally, we *reject* the null hypothesis that $\theta = \theta_0$ if we obtain a p -value that is below some conventional significance threshold (usually .05) and *fail to reject* the null hypothesis otherwise.³⁹ (Be careful, though: many tests are likely to yield many “significant” results.) This leads to a simple rule of thumb: if the coefficient is twice the size of the standard error, we can say our result is “statistically significant at the .05 level.”⁴⁰

3.4.3 The Bootstrap

The idea of using the empirical CDF to approximate a CDF motivates our final topic for this section: the bootstrap. The bootstrap provides a procedure for estimating standard errors for many statistics of interest. The bootstrap derives its name from the phrase “pull yourself up by your bootstraps.” This is an apt description, as we shall see shortly.

If we *knew* the CDF of some random variable, then we could derive the sampling distribution of any sample statistic. For example, if we knew that we had a fair coin, we would know that the sampling distribution of the sample mean would have mean $\frac{1}{2}$ and variance $\frac{1}{4n}$. In fact, we could use a computer simulation to calculate the sampling distribution of the sample mean to arbitrary

³⁹ Similarly, by convention in empirical work, if a p -value is reported without specifying θ_0 , it is assumed that the null hypothesis being tested is $\theta_0 = 0$.

⁴⁰ As a result of what is best characterized as an accident of history, this criterion ($p < 0.05$) is often considered important for publication in scientific journals. While p -values can, in some contexts, be useful metrics of the strength of evidence, a sharp delineation between “significant” and “insignificant” scientific findings based on a p -value threshold (such as $p < 0.05$) is seriously misguided—in theory and in practice.

precision by taking repeated simulated samples of size n , calculating the sample mean of each simulated sample, and observing the distribution of the resulting values. (We provide an example of such a simulation in Example 3.4.7 in the following section.)

So, characterizing the sampling variability of the sample mean (for a given n) is straightforward *if* we know the distribution that we are sampling from. The inferential problem, however, is that we do not know the distribution that we are sampling from. The bootstrap solution to this problem is quite elegant: pretend that the distribution that we are sampling from looks *exactly* like the sample that we have. That is, *plug-in* the empirical CDF for the CDF and sample from *that*. For this reason, the bootstrap is a *resampling method*.

Here, we will consider the simplest variant of the bootstrap, sometimes known as the “naive” bootstrap (for some details on alternative bootstrap procedures, see Footnote 41 and Davison and Hinkley (1997)). More specifically, to characterize the uncertainty of our estimator:

1. Take a *with replacement* sample of size n from our sample.⁴¹
2. Calculate our would-be estimate using this *bootstrap sample*.
3. Repeat steps 1 and 2 many times.
4. Using the resulting collection of *bootstrap estimates*, calculate the standard deviation of the *bootstrap distribution* of our estimator.

Given the usual plug-in regularity conditions (Section 3.3.1), this standard deviation will consistently estimate the standard error. Thus, if the asymptotic distribution of the estimator is normal, we can construct confidence intervals (and p -values) in the usual way.

This works as an inferential procedure because the bootstrap is a plug-in estimator. Under the i.i.d. sampling framework, the sampling distribution of an estimator is determined entirely by the CDF and n . Since the empirical CDF looks more and more like the CDF as n gets large, the plug-in sampling distribution yielded by the bootstrap looks more and more like the true sampling distribution. Put simply, the bootstrap works when plug-in estimation works.⁴²

The bootstrap can be used to construct confidence intervals and p -values. The most straightforward way is as part of a normal approximation using Theorems 3.4.2 (*Normal Approximation–Based Confidence Interval*) and 3.4.6 (*Normal Approximation–Based p -Values*). This approach has good theoretical

⁴¹ This is the “naive” step. Refinements to the bootstrap often replace this step with alternative assumptions (for example, in one common variant of the Bayesian bootstrap, each bootstrap sample is constructed by applying weights drawn from a prespecified distribution to our sample).

⁴² A canonical example of an estimator for which the bootstrap fails for valid asymptotic statistical inference is the sample maximum for a bounded, continuous random variable. See, e.g., Efron and Tibshirani (1994, Ch. 7.4). We will briefly discuss a practical situation in Example 7.2.4 (*Matching Estimation*) where the bootstrap may also fail.

grounding, though there are alternative ways to construct confidence intervals (and p -values) with the bootstrap, including the percentile approach. Assuming that we have B bootstrap replicates, the percentile bootstrap estimate of a 95% confidence interval can be computed by sorting the B bootstrap estimates from smallest to largest, and forming an interval by taking the $\lfloor 0.025 \times B \rfloor^{\text{th}}$ sorted bootstrap estimate and the $\lceil 0.975 \times B \rceil^{\text{th}}$ sorted bootstrap estimate. The primary advantage of the percentile approach is that it does not at any point rely on a normal approximation, and thus often has better performance when n is small. Nevertheless, for parsimony, we will rely on the naive bootstrap under a normal approximation as we proceed.

The bootstrap is particularly useful in cases where we want to conduct inference using a weird estimator, like, say, $(\hat{\beta}_0 - \hat{\beta}_1)^3 / (2 + \hat{\beta}_2^2)$. Although an analytic sampling variance estimator might be possible to derive, it might be difficult, or it might require using asymptotic approximations that yield very bad finite-sample performance. So, when we have i.i.d. sampling, the bootstrap is often a sensible way to estimate standard errors, and it saves us the hassle of having to work out an analytic sampling variance estimator in cases where it might be difficult.

3.4.4 Micronumerosity

In this section, we discuss a problem that often plagues empirical research: *micronumerosity*. As Goldberger (1991, Ch. 23.3) defined the term, micronumerosity is simply “an exotic polysyllabic name for ‘small sample size.’”⁴³ When data are *micronumerous*, we are confronted with multiple issues.

As with Goldberger (1991), to ground ideas, we will begin with the sample mean $\bar{X}$ as an estimator of the population mean $E[X]$. The primary issue is that when n is small, estimates tend to be imprecise. Theorem 3.2.34 establishes that the standard error of the sample mean is $\sigma[\bar{X}] = \sqrt{V[X]/n}$. Thus, all else equal, when n is small, the standard error of $\bar{X}$ can be very large. Accordingly, when micronumerosity is present, it is difficult to estimate features of X with much precision.

⁴³ Goldberger (1991, Ch. 23.3) coined the term in order to satirize contemporary econometric treatments of a closely related problem in regression estimation (Chapter 4) known as *multicollinearity*. (We only briefly note multicollinearity in this book.) To provide a full context for our above definition, Goldberger notes:

Econometrics texts devote many pages to the problem of multicollinearity in multiple regression, but they say little about the closely analogous problem of small sample size in estimating a univariate mean. Perhaps that imbalance is attributable to the lack of an exotic polysyllabic name for “small sample size.” If so, we can remove that impediment by introducing the term *micronumerosity*.

We are quite fond of the term, as we believe that a memorable—and indeed sometimes exotic and polysyllabic—name can help in fixing ideas.

This carries through to how we can characterize uncertainty. As our estimates are less precise, so too are our estimates of uncertainty. Since we do not know the standard error of the sample mean, we must use an estimate thereof. Under micronumerosity, the estimated standard error has high sampling variability. This issue is exacerbated when attempting to use normal approximation–based confidence intervals and p -values, which not only rely on consistency of the standard error estimator but also asymptotic normality of the sample mean. When n is small, the sampling distribution of the sample mean may not be well approximated by a normal distribution (recall Figure 3.2.2). In such circumstances, we cannot expect that normal approximation–based confidence intervals will be valid. Definitionally, asymptotically valid inferences can only be guaranteed to be valid when data are not micronumerous.

We provide an illustrative example of the problems posed by micronumerosity. Our example will also show how computer simulations can be profitably used to investigate the small-sample behavior of statistical methods.

Example 3.4.7. *Computer Simulations of the Properties of the Sample Mean under Micronumerosity*

While our analytic results (such as consistency, asymptotic normality, and asymptotic validity of confidence intervals) have been reassuring, these results tell us little when data are micronumerous. We know that the sampling distributions of $\bar{X}$ and $\hat{\sigma}[\bar{X}]$ both depend on n . Furthermore, we know that normal approximation–based confidence intervals are only guaranteed to be valid when n is very large.

To investigate the small sample properties of the sample mean and associated measures of uncertainty, we will use computer simulations. We assume that we observe n i.i.d. draws of $X \sim U(0,1)$. To perform a simulation, we simulate $n = 10$, $n = 100$, or $n = 1,000$ draws from $U(0,1)$, and compute the sample mean, estimated standard error, and 90%/95%/99% normal approximation–based confidence intervals. We perform this simulation many, many times (here, we use 1 million simulations), and collect the estimates. Due to the law of large numbers, the empirical joint distribution of these quantities from our simulations approximates the true joint distribution of the estimators arbitrarily well. Our simulations are essentially plug-in estimates of the features of a random variable (an estimator) using a large number of simulated i.i.d. draws from its distribution.

In Table 3.4.1, we present summaries of the results of our simulations. We know that the sample mean is unbiased for the population mean, so we know that $E[\bar{X}] = 0.5$ regardless of n , and we need not report the bias of the estimator. The first column, n , corresponds to the n associated with a given simulation. The second through fourth columns report properties of the standard error and our estimates thereof. The second column reports the true standard error (which can be exactly derived via Theorem 3.2.34). When $n = 10$, the variability of the estimate is quite high: the standard error

TABLE 3.4.1. *Implications of Micronumerosity (Standard Uniform Distribution)*

<i>n</i>	Standard Error Properties			Coverage		
	$\sigma[\bar{X}]$	$E[\hat{\sigma}[\bar{X}]]$	$\sigma[\hat{\sigma}[\bar{X}]]$	90% CI	95% CI	99% CI
10	0.091	0.090	0.015	86.6%	91.6%	96.5%
100	0.029	0.029	0.001	89.7%	94.7%	98.8%
1,000	0.009	0.009	0.000	90.0%	95.0%	99.0%

TABLE 3.4.2. *Implications of Micronumerosity (Bernoulli Distribution with $p = 0.5$)*

<i>n</i>	Standard Error Properties			Coverage		
	$\sigma[\bar{X}]$	$E[\hat{\sigma}[\bar{X}]]$	$\sigma[\hat{\sigma}[\bar{X}]]$	90% CI	95% CI	99% CI
10	0.158	0.157	0.014	89.0%	89.0%	97.8%
100	0.050	0.050	0.000	91.1%	94.3%	98.8%
1,000	0.016	0.016	0.000	89.3%	94.7%	99.0%

encompasses almost 10% of the range of possible values that X can take on. But as n increases, the standard error shrinks proportionally to $n^{-1/2}$. The third column reports the expected value of our estimate of the standard error, $\hat{\sigma}[\bar{X}]$. We see that, even in very small sample sizes, the estimator is approximately unbiased, as in all cases $\sigma[\bar{X}] \approx E[\hat{\sigma}[\bar{X}]]$. The fourth column shows the standard error of our estimate of the standard error; we see that our estimate of the standard error has high variability when $n = 10$, but that its variability quickly diminishes with n .

In the remaining columns, we show the actual coverage associated with our normal approximation–based confidence intervals. That is, across our many simulations, we count the number of times that the estimated confidence interval includes 0.5 and divide by the number of simulations. We see that, when n is small, our estimated confidence intervals are not valid. For example, when $n = 10$, the actual coverage of our 95% confidence intervals lies below the nominal level—they in fact have only 89% coverage. That is, across 89% of our simulated random draws of $n = 10$ values from the distribution X , the (nominally) 95% normal approximation–based confidence interval actually covers $E[X]$. However, as n gets larger, this problem is quickly alleviated. With $n = 100$, our confidence intervals appear to be very close to valid in this setting.

Of course, our calculations are specific to the setting of our simulation. The results above do not speak to all generative distributions. Table 3.4.2 reports the results of simulations conducted exactly as with Table 3.4.1, with only

one difference: X is a Bernoulli random variable with $p = 0.5$. Our results differ substantively, particularly with respect to the coverage properties of the normal approximation-based confidence intervals. We see that, even when $n = 1,000$, the 90% and 95% normal-approximation confidence intervals are not exactly valid, suggesting that some distributions require larger samples in order for asymptotic approximations to be appropriate. We also see that coverage does not necessarily follow an obvious pattern in small samples; for example, due to the differing quality of the normal approximation at different points of the distribution, the 90% confidence interval coverage follows a non-monotonic trend: the coverage is higher for $n = 100$ than for $n = 1,000$. The small-sample behavior of statistical methods justified by asymptotic theory may be somewhat unpredictable, but we can investigate some of their properties through investigation of their behavior under different assumptions about the population distribution. $\triangle$

The problems posed by micronumerosity are not limited to estimating the population mean. In fact, the problems become more pronounced when considering more complex functionals. We will face this problem shortly in Section 4.3.2 (*Overfitting*). Put simply: micronumerosity implies that we do not have enough data to estimate what we would like to estimate with enough precision. Some functionals may be easy to estimate with relatively little data, others may not. The only guaranteed solution to a problem of micronumerosity is the collection of more data as would be produced by, for instance, replication efforts.

3.5 CLUSTER SAMPLES

In this section, we consider one way in which the i.i.d. assumption might be violated. Specifically, we allow for observations to be correlated (or non-i.i.d.) within *clusters*, but i.i.d. across clusters. This accommodates a particular way in which the i.i.d. assumption might be violated, and approximates many problems that we might encounter. For example, we might believe that units within a country have correlated outcomes, but that units across countries are highly unlikely to have any statistical dependence.

To codify this idea, let us now assume that we have l i.i.d. *clusters* of observations, each consisting of m values and represented by a random vector $(X_{i1}, X_{i2}, \dots, X_{im})$. Throughout all our discussions of clustering in this book, we will be making the simplifying assumption that all clusters are of an identical size m . Largely, this serves to make matters notationally convenient and to resolve ambiguity about estimands (namely, how units across clusters should be weighted). Where this assumption is theoretically consequential, we make note of it in the text; otherwise, our results largely carry through as long as cluster sizes remain bounded. While observations must be independent across clusters (for example, $X_{ij} \perp X_{i'j}$), within clusters they may be statistically

dependent (that is, it is not generally the case that $X_{ij} \perp X_{i'j'}$). Let $n = lm$, so that the number of observations is the number of clusters times the number of observations per cluster. We now explore the consequences for estimation and inference of computing estimates from *cluster samples*.

3.5.1 Estimation with Clustering

In general, the random vector $(X_{i1}, X_{i2}, \dots, X_{im})$ is characterized by a joint CDF F . Taking l i.i.d. draws of this random vector yields an empirical joint CDF $\hat{F}$. Statistical functionals are now written as functions of the joint CDF. We can then apply the plug-in principle as we usually would for i.i.d. data, and estimate the functional of interest using the empirical joint CDF.

Let us see how this works for the sample mean. It will be useful for us to define the following random variable, the *cluster average*. The cluster average is simply the mean of the observations in that cluster.

Definition 3.5.1. Cluster Average

For i.i.d. random vectors⁴⁴ $(X_{11}, X_{12}, \dots, X_{1m})$, $(X_{21}, X_{22}, \dots, X_{2m})$, ..., $(X_{l1}, X_{l2}, \dots, X_{lm})$, the i^{th} *cluster average* is

$$W_i = \frac{1}{m} \sum_{j=1}^m X_{ij}, \forall i \in \{1, 2, \dots, l\}.$$

Since the random vectors are i.i.d., it follows that $W_1, W_2, \dots, W_l$ are i.i.d. With a single step, we have transformed the problem into an i.i.d. problem. As usual, we define the sample mean as the average of all the observed values.

Definition 3.5.2. Cluster Sample Mean

For i.i.d. random vectors $(X_{11}, X_{12}, \dots, X_{1m})$, $(X_{21}, X_{22}, \dots, X_{2m})$, ..., $(X_{l1}, X_{l2}, \dots, X_{lm})$, the *cluster sample mean* is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^l \sum_{j=1}^m X_{ij}.$$

This step is where our simplifying assumption that all clusters are the same size really becomes operational. Since clusters are all the same size, the cluster sample mean is equivalent to the sample mean of W . If we have different cluster sizes, the following theorem will not hold.

⁴⁴ Random vectors are i.i.d. if each has the same *joint* distribution and they satisfy a vector version of mutual independence.

Theorem 3.5.3. *The Cluster Sample Mean Is the Sample Mean of Cluster Averages*

Let $(X_{11}, X_{12}, \dots, X_{1m}), (X_{21}, X_{22}, \dots, X_{2m}), \dots, (X_{l1}, X_{l2}, \dots, X_{lm})$ be i.i.d. random vectors, and let W_i be the i^{th} cluster average, $\forall i \in \{1, 2, \dots, l\}$. Then

$$\bar{X} = \bar{W}.$$

Proof:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^l \sum_{j=1}^m X_{ij} = \frac{1}{lm} \sum_{i=1}^l \sum_{j=1}^m X_{ij} = \frac{1}{l} \sum_{i=1}^l \frac{1}{m} \sum_{j=1}^m X_{ij} = \frac{1}{l} \sum_{i=1}^l W_i = \bar{W}. \quad \square$$

The sample mean is now unbiased for what we will call the *population mean cluster average*, that is, the population mean of W , $E[W]$.

Theorem 3.5.4. *The Cluster Sample Mean Is Unbiased for the Population Mean Cluster Average*

Let $(X_{11}, X_{12}, \dots, X_{1m}), (X_{21}, X_{22}, \dots, X_{2m}), \dots, (X_{l1}, X_{l2}, \dots, X_{lm})$ be i.i.d. random vectors, and let W_i be the i^{th} cluster average, $\forall i \in \{1, 2, \dots, l\}$. Then

$$E[\bar{X}] = E[W].$$

Proof:

$$E[\bar{X}] = E[\bar{W}] = E[W],$$

where the second equality follows from Theorem 3.2.3 (*The Expected Value of the Sample Mean is the Population Mean*). $\square$

Characterizing the sampling variance becomes a bit more complicated, since we now need to account for the covariance between every pair of units within each cluster. However, the application of the multivariate variance rule (Theorem 2.3.2) allows for a simple representation of the sampling variance.

Theorem 3.5.5. *Sampling Variance of the Cluster Sample Mean*

Let $(X_{11}, X_{12}, \dots, X_{1m}), (X_{21}, X_{22}, \dots, X_{2m}), \dots, (X_{l1}, X_{l2}, \dots, X_{lm})$ be i.i.d. random vectors, and let W_i be the i^{th} cluster average, $\forall i \in \{1, 2, \dots, l\}$. Then the sampling variance of the cluster sample mean is

$$V[\bar{X}] = \frac{1}{lm^2} \sum_{j=1}^m \sum_{j'=1}^m \text{Cov}[X_{ij}, X_{ij'}].$$

Proof:

$$\begin{aligned}
 V[\bar{X}] &= V[\bar{W}] \\
 &= \frac{V[W]}{l} \\
 &= \frac{1}{l} V\left[\frac{1}{m} \sum_{j=1}^m X_{ij}\right] \\
 &= \frac{1}{lm^2} V\left[\sum_{j=1}^m X_{ij}\right] \\
 &= \frac{1}{lm^2} \sum_{j=1}^m \sum_{j'=1}^m \text{Cov}[X_{ij}, X_{ij'}],
 \end{aligned}$$

where the second equality follows from Theorem 3.2.4 (*Sampling Variance of the Sample Mean*) and the last follows from Theorem 2.3.2 (*Multivariate Variance Rule*). $\square$

Alternatively, we can think about the estimator as the average of i.i.d. random variables, with mean $E[X]$ and common variance $V[W]$. Accordingly, a version of the WLLN still holds, but it now says that the cluster sample mean is consistent for the population mean cluster average. This result (and the following results) will require that the number of clusters $l \rightarrow \infty$, with m not growing in n . If the number of units per cluster $m \rightarrow \infty$, no such guarantees will hold.

Theorem 3.5.6. WLLN for Cluster Samples

Let $(X_{11}, X_{12}, \dots, X_{1m})$, $(X_{21}, X_{22}, \dots, X_{2m})$, ..., $(X_{l1}, X_{l2}, \dots, X_{lm})$ be i.i.d. random vectors with finite $E[X_{ij}]$ and finite $V[X_{ij}] > 0$, $\forall i \in \{1, 2, \dots, l\}$ and $\forall j \in \{1, 2, \dots, m\}$, and let $\bar{W}_i$ be the i^{th} cluster average, $\forall i \in \{1, 2, \dots, l\}$. Then

$$\bar{X} \xrightarrow{p} E[W].$$

Proof: As m is finite, $E[W]$ and $V[W]$ are finite.⁴⁵ Then, by the WLLN, $\bar{X} = \bar{W} \xrightarrow{p} E[W]$. $\square$

⁴⁵ It could be the case that $V[W] = 0$, but then W is constant, $W = E[W]$, and so it is trivially true that $\bar{X} = \bar{W} \xrightarrow{p} E[W]$.

Similarly, the Central Limit Theorem also still applies, though the standardized sample mean no longer reduces to $\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$. Instead,

$$\begin{aligned} Z &= \frac{(\bar{X} - E[\bar{X}])}{\sigma[\bar{X}]} \\ &= \frac{(\bar{W} - E[W])}{\sigma[\bar{W}]} \\ &= \frac{(\bar{W} - E[W])}{\sqrt{V[\bar{W}]}} \\ &= \frac{(\bar{W} - E[W])}{\sqrt{\frac{V[W]}{l}}} \\ &= \frac{\sqrt{l}(\bar{W} - E[W])}{\sigma[W]} \\ &= Z_W, \end{aligned}$$

or the standardized sample mean of W .

Theorem 3.5.7. *Central Limit Theorem for Cluster Samples*

Let $(X_{11}, X_{12}, \dots, X_{1m}), (X_{21}, X_{22}, \dots, X_{2m}), \dots, (X_{l1}, X_{l2}, \dots, X_{lm})$ be i.i.d. random vectors such that $E[W]$ and $V[W] > 0$ are finite, where W_i is the i^{th} cluster average, $\forall i \in \{1, 2, \dots, l\}$. Let Z_W be the standardized sample mean. Then

$$Z_W \xrightarrow{d} N(0, 1),$$

or equivalently,

$$\sqrt{l}(\bar{X} - E[W]) \xrightarrow{d} N(0, \sigma_W^2).$$

Proof: By the Central Limit Theorem, $Z_W \xrightarrow{d} N(0, 1)$, and likewise, $\sqrt{l}(\bar{X} - E[W]) = \sqrt{l}(\bar{W} - E[W]) \xrightarrow{d} N(0, \sigma_W^2)$. $\square$

With the Central Limit Theorem in hand, we can move on to tools for inference as usual.

3.5.2 Inference with Clustering

The tools that we have used for inference can be readily adapted to the case in which we have clustered data. The sampling variance of the cluster sample mean can be estimated using a plug-in estimator:

$$\hat{V}[\bar{X}] = \hat{V}[\bar{W}] = \frac{\hat{V}[W]}{l},$$

where $\hat{V}[W]$ is the unbiased sample variance of W . By Theorem 3.2.33 (*Estimating the Sampling Variance*) and the CMT, this estimator is consistent under analogous regularity conditions to the i.i.d. case:

$$n\hat{V}[\bar{X}] = m\left(\hat{V}[\bar{W}]\right) = m\left(\hat{V}[W]\right) \xrightarrow{p} m(V[W]) = m\left(V[\bar{W}]\right) = nV[\bar{X}].$$

Since $\bar{X}$ is asymptotically normal, normal approximation-based confidence intervals and p -values can then be obtained in the usual way, except working with W instead of X .

Example 3.5.8. *Consequences of Clustering for the Sample Mean*

How does clustering affect the sampling variance of our estimates? Let us impose a working assumption for illustrative purposes. Suppose that:

- $\forall i, j, V[X_{ij}] = \sigma^2$.
- $\forall i, j, j', \rho[X_{ij}, X_{ij'}] = \rho > 0$.

In other words, the variance of each observation is identical, and every unit within a cluster shares the same correlation. Then, $\forall i, j, j', \text{Cov}[X_{ij}, X_{ij'}] = \rho[X_{ij}, X_{ij'}]\sigma[X_{ij}]\sigma[X_{ij'}] = \rho\sigma^2$, so

$$\begin{aligned} V[\bar{X}] &= \frac{1}{lm^2} \sum_{j=1}^m \sum_{j'=1}^m \text{Cov}[X_{ij}, X_{ij'}] \\ &= \frac{1}{lm^2} (m\sigma^2 + m(m-1)\rho\sigma^2) \\ &= \frac{\sigma^2}{lm} (1 + \rho(m-1)). \end{aligned}$$

Let us now consider two extreme cases (still assuming that all clusters are of an identical size m). Suppose $\rho = 0$; that is, there is no *intracluster correlation*. Then

$$V[\bar{X}] = \frac{\sigma^2}{lm} = \frac{\sigma^2}{n},$$

just as if we had no clustering. With no intracluster correlation, clustering induces no sampling variance inflation, so it is as though we have n i.i.d. observations.

Now suppose $\rho = 1$; that is, there is perfect intraclass correlation. Then

$$V[\bar{X}] = \frac{\sigma^2}{lm}(1 + m - 1) = \frac{\sigma^2}{lm}m = \frac{\sigma^2}{l}.$$

With perfect intraclass correlation, it is as though we only have l observations. Usually, the answer lies somewhere in between. The primary consequence of clustering is that we have less information from the data, and we suffer a precision penalty accordingly. Even if n is very large, if l is small, the data may nevertheless effectively be micronumerous. Δ

What can we do if we want to conduct inference using an arbitrary estimator and we do not have (or do not want to derive) an analytic estimator of the sampling variance of that estimator? We again can use a variant of the bootstrap. Specifically, with clustering, one appropriate method is the *block bootstrap*, which is exactly like the bootstrap except that now we resample whole *clusters* rather than individual units. To implement the block bootstrap:

1. Take a *with replacement* sample of m clusters from our sample, keeping clusters intact.
2. Calculate our would-be estimate using this collected n observations in our block bootstrap sample.
3. Repeat steps 1 and 2 many times.
4. Using the resulting collection of block bootstrap estimates, calculate the standard deviation of the block bootstrap distribution of our estimator.

Since we drew a sample of l i.i.d. clusters to begin with, we are again replicating the sampling process, taking with-replacement samples of l clusters from our sample. The rest of the procedure is the same as before. The block bootstrap behaves well, as usual, given a large number of clusters and the usual plug-in regularity conditions.

3.6 FURTHER READINGS

Again, the readings in Chapter 1 and 2 remain relevant. Beyond these, there are a number of detailed treatments of the topics discussed in Chapter 3. For a general treatment of statistical estimation theory, we strongly recommend Wasserman (2006) and Wasserman (2012), which serve as a key inspiration for the presentation of the materials in this chapter. For the theory of survey samples, Cochran (1977) is a canonical treatment. Although we do not cover design-based inference in our book, Särndal et al. (1992) and Thompson (1997) provide excellent modern treatments of this approach to survey sampling. Lehmann (1999) and DasGupta (2008) are technical and invaluable resources for asymptotic theory. Tsybakov (2009) provides an elegant, technical treatment of nonparametric estimation theory. Newey (1990) provides an article-length overview of semiparametric efficiency theory, and remains

perhaps the most readable (yet still technical and rigorous) treatment of the subject matter. Van der Laan and Rose (2011) provides a modern book-length unification of semiparametric methods and machine learning for the estimation of statistical functionals. Silverman (1986) provides an in-depth book-length treatment of kernel density estimation. Greenland et al. (2016) is a concise, important, and highly readable article about statistical testing and confidence intervals, and is strongly suggested for all readers. Lehmann and Romano (2005) also covers the theory of statistical tests, which covers much of the same material as Lehmann (1999), but at a higher level of technicality. Efron and Tibshirani (1994) and Davison and Hinkley (1997) are quite readable and comprehensive books on the bootstrap.